

HEART-BENCH: Do LLMs Make Persona-Consistent Decisions?

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Abstract

While LLMs have demonstrated strong task-oriented capabilities, their ability to make persona-consistent decisions grounded in long-term memory remains underexplored. We introduce HEART-BENCH, a benchmark of psychology-informed, memory-grounded persona consistency in controlled synthetic settings. HEART-BENCH defines 11 characters using Big Five–anchored profiles and value-based specifications, assigns each character 1,000 synthetic autobiographical memories, and evaluates them across 64 DIAMONDS-guided decision scenarios. Each character–scenario pair is associated with a response that the character is expected to have under the specific scenario. Three domain experts independently compare anonymized candidates, and a senior expert selects, revises, or rewrites the final reference when needed. This process yields 673 multiple-choice questions in which a model selects the option judged most consistent with the predefined character and its history. A separate expert panel audits 176 final items, all of which receive majority support for the existing key. We evaluate 12 LLMs under a No-Retrieval baseline and three memory-augmented settings: Naive-RAG, Mem0, and PersonaDB. The strongest configuration, Gemini-3.1-Pro with Naive-RAG, reaches 63.3% accuracy, while most models remain below 40%. These results show substantial difficulty in selecting expert-adjudicated responses consistent with predefined characters and their memories.

1 Introduction

Large language models (LLMs) have achieved substantial progress in planning, reasoning, and action (Anthropic, 2026b; OpenAI, 2026b; Google DeepMind, 2026). As LLM agents are increasingly deployed in entertainment (Chen et al., 2024; Tseng et al., 2024), AI companionship (Maples

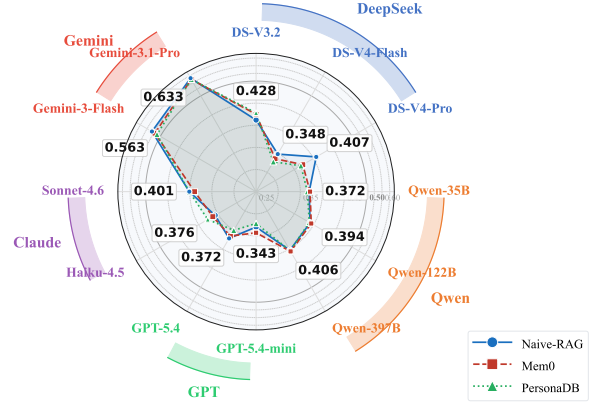


Figure 1: Accuracy for all 12 models, shown separately for the three memory-augmented settings.

et al., 2024; Phang et al., 2025), and persona-cloning systems (Wei et al., 2025; Lee et al., 2026), users increasingly expect persistent and personalized behavior. These applications require more than conversational fluency and memory recall: a system should use a character’s traits, values, and prior experiences consistently when choosing how to respond. Existing evaluations, however, primarily target dialogue, explicit reasoning, or memory retrieval. This leaves a narrower but important question unresolved: *Given a predefined character, its history, and a scenario, can an LLM select a persona-consistent response?*

To address this gap, we introduce HEART-BENCH, a controlled benchmark for evaluating memory-grounded persona consistency. HEART-BENCH constructs 11 synthetic characters grounded in Big Five–anchored profiles, each associated with 1,000 structured synthetic autobiographical memories (Singer, 1995) organized into eight age-defined life-stage bins informed by developmental theories. We further design 64 decision scenarios guided by the DIAMONDS framework (Rauthmann et al., 2014). Each character–scenario pair is associated with a

Table 1: Comparison of HEART-BENCH with related benchmarks across six dimensions (see Appendix A for dimension definitions). ✓ yes; ✗ no; ~ partial.

Benchmark	Big Five-grounded	Age-defined life-stage coverage	Episodic memory	Scenario eval.	Expert annot.	Persona-consistent decision
CharacterEval (Tu et al., 2024)	✗	✗	✗	✓	~	✗
PersonaHub (Ge et al., 2024)	✗	✗	✗	✓	✗	✗
LoCoMo (Maharana et al., 2024)	✗	✗	✓	✓	✗	✗
CloneMem (Hu et al., 2026)	✓	~	✓	✗	✗	✗
PersonaLens (Zhao et al., 2025)	✗	✗	~	✓	~	✗
PersonaMem (Jiang et al., 2025)	✗	✗	~	✓	✗	✗
HEART-BENCH	✓	✓	✓	✓	✓	✓

response that the character is expected to have under the specific scenario. To label ground truth responses, three domain experts independently compare candidate responses, and a senior expert handles every case not unanimously accepted as-is by selecting, revising, rewriting, or discarding the response. The adjudicated character response becomes the keyed reference, while responses associated with other characters in the same scenario provide distractor candidates. This process yields 673 multiple-choice questions; the memory-augmented settings test whether models can use relevant synthetic memories to identify the keyed response. Table 1 situates HEART-BENCH relative to representative persona and memory benchmarks across six design dimensions.

We evaluate 12 LLMs from the GPT, Claude, Gemini, DeepSeek, and Qwen families under a No-Retrieval baseline and three memory-augmented settings: Naive-RAG (Lewis et al., 2020), Mem0 (Chhikara et al., 2025), and PersonaDB (Sun et al., 2025). The strongest configuration, Gemini-3.1-Pro with Naive-RAG, reaches 63.3% accuracy, while most models remain below 40%. Within this controlled benchmark, the evaluated models therefore show limited ability to use encoded character and memory cues to select the expert-adjudicated response.

We summarize our contributions as follows:

- We introduce a controlled benchmark comprising 11 psychology-informed synthetic characters, 11,000 synthetic autobiographical memories, 64 decision scenarios, and 673 expert-adjudicated MCQs.
- We operationalize memory-grounded persona consistency as agreement with an expert-adjudicated reference response conditioned on a predefined character, history, and scenario, and independently audit 176 items for reference-label reliability.

- We evaluate 12 LLMs across No-Retrieval and three memory-augmented settings, quantifying their ability to use encoded character and memory cues within this controlled synthetic setting.

2 HEART-BENCH

2.1 Overview

Figure 2 illustrates the full pipeline of HEART-BENCH. The benchmark comprises two components—an 11-character set informed by the Big Five (McCrae and John, 1992) and Schwartz’s value theory (Schwartz, 1992), with each character equipped with 1,000 synthetic long-horizon memories, and a *scenario suite* of 64 character-agnostic situations that span the eight DIAMONDS (Rauthmann et al., 2014) dimensions—Duty, Intellect, Adversity, Mating, pOsitivity, Negativity, Deception, and Sociality—and are distributed across the eight age-defined life-stage bins. For each retained character–scenario pair, experts review candidate responses and adjudicate the response judged most consistent with the predefined profile and history. The resulting reference responses are assembled into MCQs in which the target character’s adjudicated response is keyed and responses associated with other characters serve as distractor candidates. After expert review, adjudication, and filtering, HEART-BENCH contains 673 evaluation instances. Across the corpus, each character’s 1,000 memories total ~1.4M tokens; the headline statistics are consolidated in Table 9 (Appendix B.7).

2.2 Character Construction

We define a synthetic character as $c = (\mathbf{p}, \{m_i\}_{i=1}^T)$, where $\mathbf{p} = (O, C, E, A, N) \in [0, 1]^5$ is a predefined Big Five vector and $\{m_i\}_{i=1}^T$ is a synthetic autobiographical history informed by

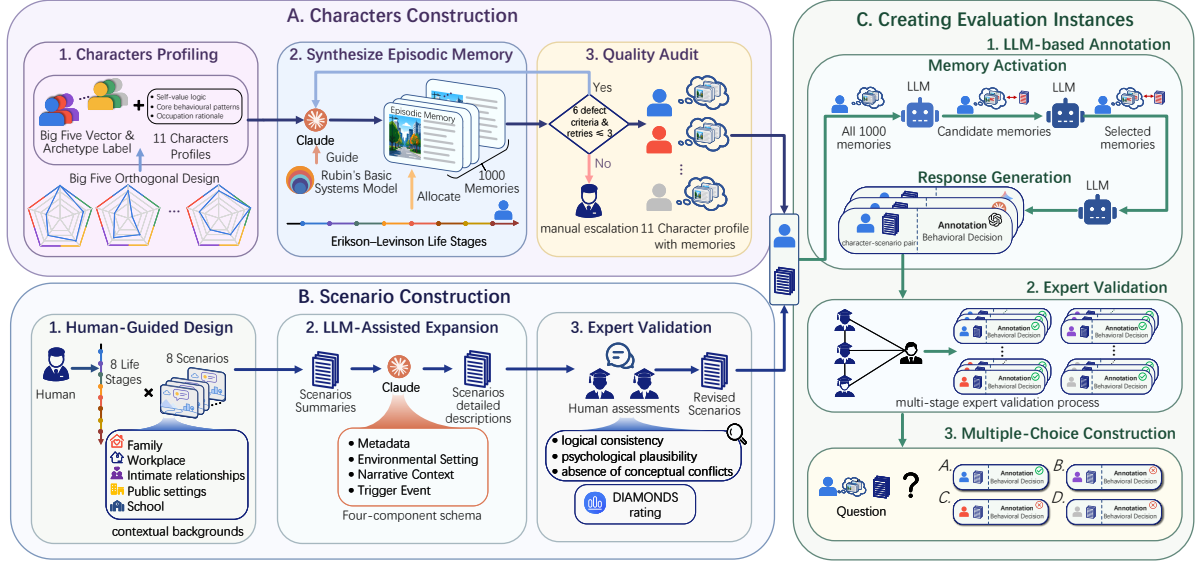


Figure 2: The pipeline has four stages: character construction, scenario construction, expert review and adjudication of reference responses, and conversion into MCQ format for model evaluation.

research on narrative memory (Singer, 1995). We construct each character through three stages.

Stage 1: Character Profiling. We anchor each character to the Big Five factor model (McCrae and John, 1992) rather than using a loosely defined role description. Our trait-anchored coverage design assigns one defining dimension to each profile: for every OCEAN trait, we instantiate one high ($p = 0.95$) and one low ($p = 0.10$) anchor, together with an all-neutral ($p = 0.50$) baseline, yielding 11 coherent multi-trait profiles (Table 4). For each OCEAN trait, the high and low anchors differ by 0.85 on the defining dimension, and each anchor differs from the neutral baseline by at least 0.40 on that dimension. Non-anchor dimensions vary to preserve nonuniform multi-trait character profiles. Each profile is augmented with Schwartz values derived from established trait-value correlations (Schwartz, 1992; Roccas et al., 2002), an occupation, a self-value logic, and eight behavioral patterns that guide memory generation (Appendices B.1 and B.2). The resulting profiles were reviewed by the same four-expert team that adjudicated the reference responses. This design covers both poles of every trait; differences are interpreted at the character level rather than as causal effects of an isolated trait.

Stage 2: Synthesizing Autobiographical Memories. Claude Sonnet 4.6 generates 1,000 episodic memories per character from ages 6–50. Memories are distributed across eight age-

defined life-stage bins informed by Erikson (Erikson, 1963) and Levinson (Levinson et al., 1978; Levinson, 1986), with higher assigned density in adolescence and early-adult bins. The 45 one-year bins also emphasize the age-30 interval. As illustrated in Figure 3, each narrative integrates five operational dimensions inspired by Rubin’s Basic Systems Model (Rubin, 2006)—sensory detail, dialogue reconstruction, inner monologue, somatic response, and aftermath—and is diversified across workplace, interpersonal, health, creative, and daily-life domains. This yields a longitudinally organized synthetic history with rich episodic content. The allocation scheme, generation protocol, and complete prompt are provided in Appendices B.3, B.4, and J.3.

Stage 3: Quality Audit. Each memory is screened against six automated defect criteria covering perspective drift, age-inappropriate retrospection, metadata leakage, duplication, length, and schema compliance. Failures enter a targeted repair loop; memories passing these checks are then deduplicated, chronologically ordered into a 1,000-entry timeline, and sanitized with gender-neutral labels. Full criteria and repair procedures appear in Appendix B.5, with the final schema and statistics in Appendix B.7.

2.3 Scenario Construction

A scenario defines a character-agnostic decision context that serves as a shared external stimulus in our evaluation (see Appendix C.1 for a complete

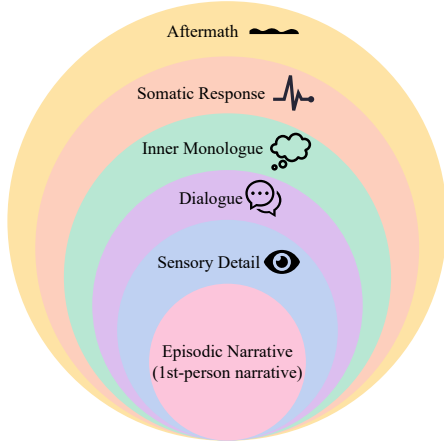


Figure 3: Five operational dimensions, inspired by Rubin’s Basic Systems Model (Rubin, 2006), used to structure synthetic episodic narratives.

example). Each compatible character is placed in the same environmental and narrative context and required to respond to a decision-triggering event. During evaluation, every LLM receives the scenario, options, character ID, and occupation; memory-augmented settings additionally provide retrieved synthetic memories. The model must identify the response keyed to the target character. By holding the external situation fixed across characters, scenarios enable controlled comparison of whether LLMs can distinguish response differences encoded in the predefined profiles and synthetic histories.

Formally, a scenario is represented as $s = \langle m, \mathcal{C}_{\text{env}}, \mathcal{C}_{\text{narrative}}, e_{\text{trigger}} \rangle$, where m contains the scenario metadata, including its target life-stage bin and intensity; $\mathcal{C}_{\text{env}}$ and $\mathcal{C}_{\text{narrative}}$ specify the environmental and narrative contexts, respectively; and e_{trigger} denotes the event that prompts an immediate response.

We construct the scenario suite through a three-stage human-in-the-loop pipeline: expert blueprinting, LLM-assisted expansion, and expert review.

Stage 1: Human-Guided Blueprinting. To ensure structural diversity, clinical experts manually draft the foundational summaries for all scenarios, using the DIAMONDS (Rauthmann et al., 2014) framework to structure coverage of situational dimensions. We design exactly 8 scenarios per life-stage bin, yielding 64 scenarios across eight bins. These summaries cover family, school, workplace, relationship, and public-space contexts. Each summary establishes a core tension while maintaining

the character-agnostic constraint.

Stage 2: LLM-Assisted Expansion. We use Claude Opus 4.6 (Anthropic, 2026a) to expand the expert-drafted summaries into structured scenario descriptions based on a four-component schema (metadata, environmental setting, narrative context, and trigger event), enforcing consistent fields across age metadata and context. The full schema and a complete worked example are given in Appendix C.1.

Stage 3: Expert Review. Two doctoral-level psychology researchers review the generated scenarios for logical plausibility and assess their coverage using two theory-informed lenses: DIAMONDS (Rauthmann et al., 2014) ratings describe the eight situational dimensions at varying intensity levels, and Schwartz’s value circumplex (Schwartz, 1992) is used to check intended value-conflict alignment. The resulting rating and conflict distributions document coverage under this designed scenario set (protocol and full distributions in Appendix C.2).

2.4 Creating Evaluation Instances

Having built the character and scenario sets, we formulate each test instance as an MCQ whose keyed reference is the response judged by experts to be most consistent with the predefined target character, its synthetic history, and the scenario. This adjudicated response serves as the benchmark label.

Stage 1: Automated Annotation with LLMs. For each (character, scenario) pair, three parallel family-specific pipelines (Claude, GPT, and Gemini) retrieve relevant episodic memories and generate candidate responses conditioned on activated memories and the character’s full profile (Big Five traits, Schwartz values, and self-narrative). Using multiple model families reduces dependence on any single annotation model and provides diverse candidates for expert comparison. Candidate order is randomized and model-family identities remain hidden throughout review. Full model configurations, prompts, and an annotation example are provided in Appendix D.1.

Stage 2: Expert Review and Adjudication. Three primary annotators, each with 5+ years of personality-assessment experience, independently assess pair suitability and compare anonymized candidates; a professor with expertise in psychology and computer science serves as arbitrator. Under a consensus-then-arbitration rule, a candidate

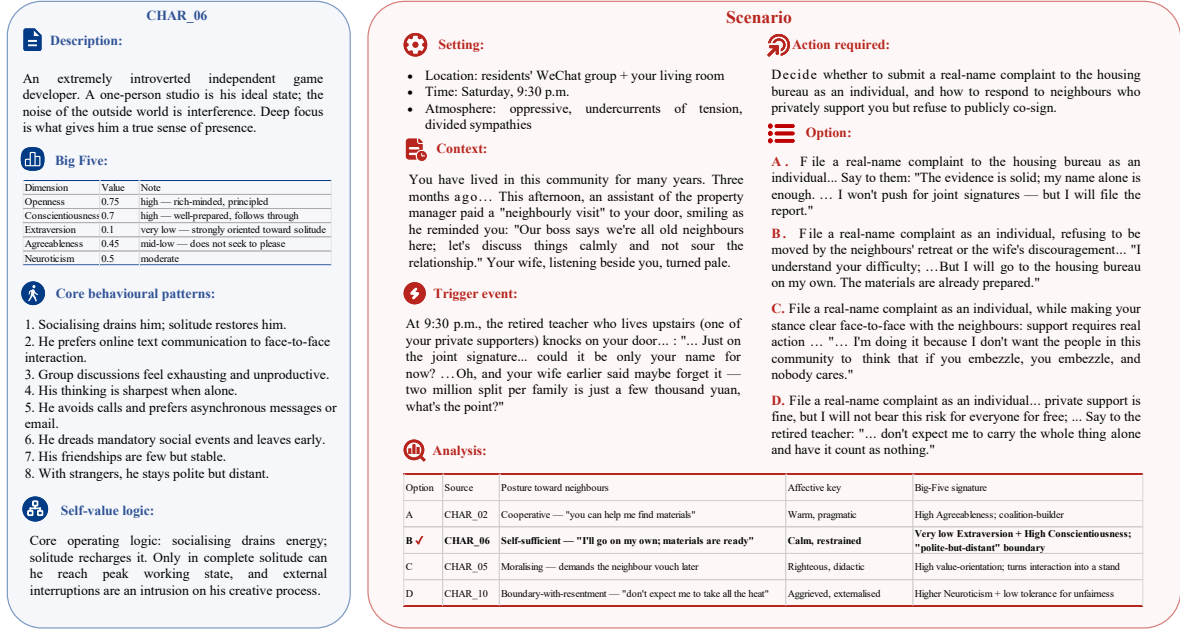


Figure 4: A HEART-BENCH instance requiring profile-response alignment beyond surface action matching; detailed analysis is in Appendix D.5.

is adopted directly only when all three annotators select it as *acceptable as-is*; otherwise, the arbitrator selects, edits, or rewrites the response, or discards an unsuitable pair. This process removed 31 pairs and retained 673 expert-adjudicated reference responses. Experts changed approximately 15% of the words in selected LLM candidates on average, documenting substantive human involvement in constructing the references. Full guidelines and outcome statistics are provided in Appendix D.2.

Stage 3: Test Instance Assembly. For each retained annotation, the target character's adjudicated response becomes the keyed reference option, while *Claude-Sonnet-4.6* (Anthropic, 2026c) selects three distractors from the available adjudicated responses associated with other characters in the same scenario. The selection excludes responses that are either too similar to the reference or obviously irrelevant, avoiding ambiguous or trivial questions. The four options are randomly shuffled to remove positional bias, yielding 673 MCQs. Full assembly rules are provided in Appendix D.3.

After assembly, three experts independent of the original construction team assessed a stratified random subset of 176 final MCQs. All three experts supported the existing reference option on 139 items (79.0%), while two of three supported it on

the remaining 37 items (21.0%); thus, every sampled item received majority support for the existing key. Pre-adjudication inter-rater reliability was high (Fleiss' $\kappa = 0.81$; Krippendorff's $\alpha = 0.80$). All 37 items without complete agreement entered adjudication, and the adjudicator confirmed that each item and keyed answer was usable as-is. No wording edits or keyed-answer changes were required, and the audit did not alter the evaluated dataset or primary scores. These results support reference-label reliability on the audited subset. The full protocol and results are provided in Appendix D.4.

2.5 Example of HEART-BENCH

Figure 4 illustrates profile-response alignment beyond surface action matching. CHAR_06 combines very low Extraversion (0.10), high Conscientiousness (0.70), and a "polite-but-distant" interaction pattern. Option A preserves the complaint action but adds coalition building, whereas Options C and D introduce moralizing and blame. Experts keyed Option B because its prior preparation, individual action, and concise low-contact wording best match the encoded profile. The example shows how shared actions can differ in persona-specific pragmatic framing; Appendix D.5 provides a fuller analysis.

Table 2: Accuracy on HEART-BENCH. Memory-Aug. Avg. averages the three retrieval-based settings, excluding No-Retrieval. Gemini-3.1-Pro performs best. Scenario-clustered 95% CIs are in Appendix G.

Model	No-Retrieval LLM	Naive-RAG	Mem0	PersonaDB	Memory-Aug. Avg.
<i>Open-source</i>					
DeepSeek-V3.2	0.285	0.412	0.425	0.428	0.422
DeepSeek-V4-Flash	0.278	0.348	0.336	0.328	0.337
DeepSeek-V4-Pro	0.291	0.407	0.372	0.367	0.382
Qwen3.5-35B-A3B	0.254	0.370	0.372	0.366	0.369
Qwen3.5-122B-A10B	0.264	0.391	0.394	0.389	0.391
Qwen3.5-397B-A17B	0.275	0.403	0.406	0.404	0.404
<i>Closed-source</i>					
GPT-5.4-mini	0.288	0.330	0.343	0.322	0.332
GPT-5.4	0.278	0.372	0.366	0.351	0.363
Claude-Haiku-4.5	0.245	0.357	0.363	0.376	0.365
Claude-Sonnet-4.6	0.257	0.401	0.389	0.401	0.397
Gemini-3-Flash	0.253	0.563	0.541	0.525	0.543
Gemini-3.1-Pro	0.264	0.633	0.624	0.624	0.627

3 Experiments

3.1 Experimental Setup

Model Selection. We evaluate 12 representative LLMs, including six open-source models: *DeepSeek-V3.2* (DeepSeek-AI, 2025), *DeepSeek-V4-Flash*, *DeepSeek-V4-Pro* (DeepSeek-AI, 2026), *Qwen3.5-35B-A3B*, *Qwen3.5-122B-A10B*, and *Qwen3.5-397B-A17B* (Qwen Team, 2026); and six closed-source models: *GPT-5.4-mini*, *GPT-5.4* (OpenAI, 2026a), *Claude-Haiku-4.5* (Anthropic, 2025), *Claude-Sonnet-4.6* (Anthropic, 2026c), *Gemini-3-Flash* (Google DeepMind, 2025), and *Gemini-3.1-Pro* (Google DeepMind, 2026). All models are evaluated under a unified, deterministic decoding protocol (Appendix F).

Memory-Augmented Settings. We compare three retrieval-based frameworks—Naive-RAG (Lewis et al., 2020), Mem0 (Chhikara et al., 2025), and PersonaDB (Sun et al., 2025)—against a *No-Retrieval LLM* baseline. During evaluation, all models receive the scenario, answer options, character ID, and occupation. For all three memory-augmented settings, both the memory backend and the model-facing prompt use only the de-identified `content_full` narrative; no other stored memory field enters the evaluation pipeline. All retrieval frameworks use the same embedding and auxiliary memory-management models, with retrieval budgets calibrated to provide comparable episodic content. The underlying Big Five vector, value logic, behavioral patterns, and structured psychological annotations are withheld to prevent direct reference-label leakage. Full implemen-

tation and field-isolation details are provided in Appendix E; the model-facing evaluation template is reproduced in Appendix J.1.

Evaluation Metrics. We report *accuracy*, defined as the proportion of the 673 MCQs for which a model selects the expert-adjudicated keyed option. Only the model’s `decision_choice` field is scored; the accompanying free-text rationale fields are not compared with the reference response and do not enter the metric. This metric quantifies agreement with HEART-BENCH references. Statistical procedures are detailed in Appendix G. Reference construction uses richer profile information than evaluation. Accuracy therefore jointly reflects retrieval coverage, recovery of persona cues, and agreement with expert-adjudicated references; it should not be interpreted as decision quality given an explicit profile (Appendix E).

3.2 Results and Discussion

Model choice dominates within the benchmark. Under Naive-RAG, the spread between the best- and worst-performing models exceeds 30% (63.3% vs. 33.0%), whereas the average gap between retrieval methods for a given model is only 1.7% and never exceeds 4%. The model- and method-axis Gini mean differences are 0.088 and 0.006, respectively, yielding a difference of 0.082 (95% CI 0.068–0.096). Gemini-3.1-Pro achieves the best accuracy (63.3%, 95% CI 59.2–67.4%) and ranks first by Memory-Augmented Avg. in all 10,000 scenario-bootstrap resamples. All non-Gemini models achieve only 33.2%–

Table 3: Memory-content ablation on the 176-item subset. Accuracy is averaged over the 12 LLMs.

Cond.	Character source	Memory selection	Acc.
C0	Target	Target scenario	0.399
C1	Target	Other same-bin scenario	0.349
C2	Other	Target scenario	0.227
C3	Target	Random non-target	0.345

42.2% Memory-Augmented Avg., showing limited agreement with the expert-adjudicated references for synthetic characters under this setup.

Retrieval helps, with no significant differences among backends. Retrieval outperforms No-Retrieval in all 36 Holm-corrected tests, whereas no comparison among the three retrieval backends is significant. Rankings remain strongly correlated across backends ($\rho = 0.89\text{--}0.94$), and Mem0 and PersonaDB have lower point estimates than Naive-RAG for 6 and 8 of the 12 models, respectively. Their additional summarization and structural abstraction may obscure latent character cues in the original episodic memories, although the nonsignificant backend comparisons do not establish equivalence.

Memory ownership and scenario alignment are both informative. On a fixed 176-item subset, we evaluate the same 12 models under Naive-RAG with $k=30$ while varying only the supplied memory assignment: C0 uses target-scenario memories from the target character; C1 uses memories retrieved for another scenario in the same life-stage bin from the target character; C2 uses target-scenario memories from another character; and C3 uses randomly sampled eligible non-target memories from the target character. The scenario, answer options, model, decoding protocol, and number of supplied memories are held fixed; no condition exposes the latent Big Five vector, Schwartz profile, self-value logic, or behavioral templates. C0 achieves 39.9% accuracy, exceeding C1, C2, and C3 by 5.06, 17.24, and 5.43 percentage points, respectively, with all 12 models favoring C0 in each contrast. Across paired model units, all three C0 contrasts remain significant under two-sided exact sign-flip tests after Holm correction over six comparisons (adjusted $p=0.00293$ for each). Thus, scenario-matched, character-owned memories provide information beyond either arbitrary same-character memories or scenario-matched memories from another char-

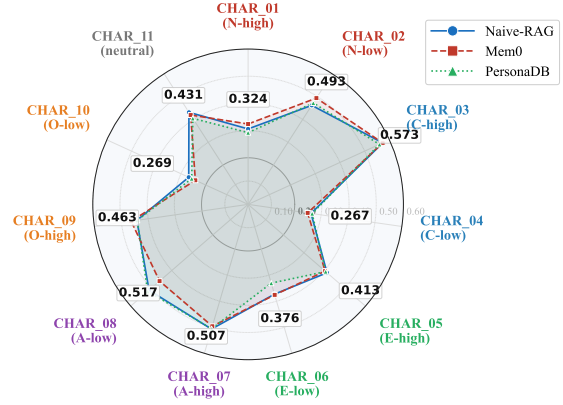


Figure 5: Per-character accuracy averaged across LLMs by memory-augmented setting.

acter. Full protocol and per-model results are provided in Appendix H.

Gemini’s lead persists across reference-response provenance subsets. Because Gemini-origin candidates account for 497 of the 673 adjudicated references, same-family annotation bias is a natural concern. Gemini-3.1-Pro and Gemini-3-Flash nevertheless remain the top two models by Memory-Augmented Avg. on the Gemini-origin (63.3% and 54.2%), Claude-origin (61.6% and 56.1%), and GPT-origin subsets (59.3% and 50.0%), with source-specific ranking correlations of $\rho = 0.666\text{--}0.809$. On non-Gemini-origin items, the Gemini family retains a 17.3-point advantage. Provenance interactions are nonsignificant in both scenario-clustered difference-in-differences (4.9 points, 95% CI $-1.1\text{--}10.8$, adjusted $p=0.3395$) and item-fixed-effect analyses (5.4 points, 95% CI $-0.6\text{--}11.5$, adjusted $p=0.2759$). These analyses do not detect a statistically significant family-specific provenance interaction, although they do not establish the absence of source bias (Appendix I.1).

Character-level difficulty and error patterns. Figure 5 shows substantial variation across characters: C-High (CHAR_03) reaches 57.3%, whereas C-Low (CHAR_04, 26.7%) and O-Low (CHAR_10, 26.9%) are the two hardest characters. N-High (CHAR_01, 32.4%) is also among the more difficult characters. Errors are concentrated: among 221 persistent shared-failure items, 84.2% have at least half of the Naive-RAG models converging on the same non-reference option. Dominant confusions include C-Low→C-High, N-High→N-Low, and O-Low→O-High, indicating

structured opposite-pole errors rather than a uniform distribution.

4 Related Work

4.1 Persona Modeling in LLM Systems

The Big Five (OCEAN) framework (McCrae and John, 1992) is a prominent account of stable individual differences, with predictive validity across cultures (McCrae and Costa, 1997) and the lifespan (Roberts et al., 2007). Human-personality research also describes behavioral states that vary with situational demands (Fleeson, 2001), autobiographical experiences organized around self-relevant goals (Conway and Pleydell-Pearce, 2000), and developmental schemas associated with later reactions (Young et al., 2006). These theories motivate the structured synthetic artifacts in our benchmark.

Current LLM persona systems follow two broad lines. The first injects static role descriptions or trait vectors at inference time, as in CharacterLLM (Shao et al., 2023), PersonaLLM (Jiang et al., 2024), PersonaAgent (Zhang et al., 2025a), and RGMem (Tian et al., 2025). The second develops long-term memory infrastructure, including MemGPT (Packer et al., 2023), HippoRAG (Gutiérrez et al., 2024), Mem0 (Chhikara et al., 2025), and A-MEM (Xu et al., 2025). Existing systems usually provide either static profiles or long-term memory stores; few benchmarks test response selection conditioned jointly on a predefined trait/value profile and its synthetic memory history.

4.2 Benchmarks for Persona- and Psychology-Informed Evaluation

Psychometric evaluation of LLMs. Prompted LLMs can simulate Big Five questionnaire responses with reasonable reliability (Serapio-García et al., 2023), and related work uses MBTI or Big Five scales to characterize model behavior. However, these approaches treat psychometric instruments as stimuli rather than testing persona-consistent decisions in controlled scenarios. They also evaluate context-free snapshots rather than response selection conditioned jointly on a profile and a structured synthetic history (McAdams, 2001; Pasupathi et al., 2007).

Persona and memory benchmarks. Conversational persona benchmarks generally test factual or stylistic consistency. Long-horizon benchmarks

such as LoCoMo (Maharana et al., 2024) and LongMemEval (Wu et al., 2024) extend evaluation to multi-session dialogue but primarily assess recall. CloneMem (Hu et al., 2026), RealMem (Bian et al., 2026), and KnowMeBench (Wu et al., 2026) use richer personal evidence, yet their central targets remain recall, preference inference, or goal consistency. In contrast, HEART-BENCH evaluates memory-grounded persona-consistent response selection under controlled scenarios structured by DIAMONDS (Rauthmann et al., 2014).

Data substrates and evaluation depth. Synthetic chat logs and task-oriented interactions often lack the perceptual, affective, and introspective detail represented in autobiographical memories (Rubin, 2006), making memory-grounded character reasoning difficult to distinguish from surface pattern matching. Existing benchmarks also rarely combine content-rich episodic narratives with explicitly age-indexed synthetic histories (McAdams, 2001). HEART-BENCH addresses this gap by using dimensions inspired by Rubin’s Basic Systems Model (Rubin, 2006) to structure generated memories and organizing them into eight age-defined life-stage bins informed by developmental theories (Erikson, 1963; Levinson et al., 1978).

5 Conclusion

In this work, we presented HEART-BENCH, a controlled benchmark for memory-grounded, persona-consistent response selection in LLMs. It combines 11 predefined synthetic characters, 11,000 synthetic autobiographical memories, 64 shared decision scenarios, and 673 MCQs keyed by expert-adjudicated reference responses. Across 12 models and four evaluation settings, agreement with these references remains limited; the memory-content ablation further shows higher accuracy when supplied memories match both the target character and the scenario. The shared-scenario structure supports controlled comparisons of retrieval, memory representation, and persona-conditioned response selection methods. We intend HEART-BENCH as a testbed for persona consistency and memory grounding.

Limitations

The central limitation is construct scope: HEART-BENCH evaluates agreement with

expert-adjudicated references for synthetic characters rather than behavior observed from real people. Expert review supports content plausibility and reference-label reliability. A single keyed option compresses potentially plural, context-dependent behavior, and the static MCQ format does not measure interactive agent behavior. Because several evaluated model families also contributed candidate responses, source-family or stylistic cues may remain despite anonymized expert adjudication and our provenance-stratified analysis.

The benchmark is also limited to 11 characters, 64 scenarios, and single-turn choices, restricting coverage of cultures, demographics, naturalistic interaction, and within-person change. Its expert-intensive construction limits scale, and the independent audit covers 176 of 673 items, supporting a dataset-level reliability estimate rather than item-wise validation of the full benchmark. Future work should incorporate human-authored candidates, distributions over plausible responses, multi-turn settings, and references derived from observed behavior to evaluate external validity.

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A Benchmark Comparison Dimensions

Table 1 in the main text compares HEART-BENCH with related benchmarks across six dimensions. We define each dimension below. These dimensions do not exhaust all aspects of character and memory evaluation; they operationalize design properties relevant to psychology-informed, memory-grounded persona-consistent response selection in controlled scenarios.

- **Big Five-grounded.** The benchmark explicitly adopts the Big Five (OCEAN) personality framework as the basis for character design—not by merely mentioning personality, but by operationalizing it through structured trait dimensions.
- **Age-defined life-stage coverage.** The benchmark organizes synthetic memories into age-defined life-stage bins (e.g., childhood, adolescence, adulthood) informed by developmental theories (e.g., Erikson, Levinson). This criterion captures the temporal organization of synthetic memories.
- **Episodic memory.** The benchmark includes or evaluates structured synthetic autobiographical/episodic content—first-person narratives representing events, affective cues, and intended consequences for the character—rather than mere interaction logs or fact stores.
- **Scenario eval.** Evaluation requires a model to select a concrete response within a structured situational context. Pure factual recall, question answering, or stylistic consistency checks do not qualify.
- **Expert annot.** The benchmark’s reference labels or quality judgments involve domain experts (e.g., psychologists, clinical researchers) as primary annotators or reviewers—not crowd workers or automated scoring alone.
- **Persona-consistent decision.** Evaluation specifically tests whether selected responses are consistent with the predefined character profile and history. Benchmarks that only test

Table 4: The 11 character archetypes. Bolded scores mark the defining extreme dimension; CHAR_11 is the all-neutral baseline.

ID	Archetype	O	C	E	A	N	Occupation
CHAR_01	N-High	.85	.50	.30	.55	.95	Freelance content creator
CHAR_02	N-Low	.50	.55	.50	.60	.10	Community GP
CHAR_03	C-High	.45	.95	.40	.60	.30	Project manager
CHAR_04	C-Low	.80	.10	.60	.65	.45	Freelance programmer
CHAR_05	E-High	.65	.55	.95	.70	.35	Sales director
CHAR_06	E-Low	.75	.70	.10	.45	.50	Indie game developer
CHAR_07	A-High	.55	.60	.55	.95	.55	Non-profit coordinator
CHAR_08	A-Low	.65	.85	.45	.10	.35	Startup co-founder
CHAR_09	O-High	.95	.35	.65	.55	.40	Anthropology lecturer
CHAR_10	O-Low	.10	.75	.40	.55	.45	Building-materials retailer
CHAR_11	Neutral	.50	.50	.50	.50	.50	Local journalist

whether a model recalls facts or matches a surface stylistic pattern do not qualify.

B Character Construction and Memory Pipeline

This appendix expands the three-stage character-construction pipeline described in the main text (§2.2): profile specification, autobiographical-memory synthesis, and quality audit. For implementation detail, the synthesis stage is decomposed below into memory-quota allocation and memory generation; we also report the unified memory schema and resulting dataset statistics.

B.1 Profile Specification

Each of the 11 character profiles is stored as a JSON object in `characters_phase11.json`. Beyond the Big Five vector and archetype label reported in Table 4, each profile contains:

- **Self-value logic** (1 sentence): the character’s core cognitive operating principle, derived from the dominant trait.
- **Core behavioral patterns** (8 items): fixed trait expressions that must appear across all generated memories.
- **Occupation rationale**: the occupation is chosen to be consistent with the dominant trait so that work-domain memories are ecologically valid.

Table 5 provides the self-value logic and three representative core behavioral patterns for each character. Full 8-pattern lists are stored in `characters_phase11.json`.

Table 5: Character specifications: self-value logic and representative behavioral patterns.

ID	Self-Value Logic	Representative Core Patterns (of 8)
CHAR_01 N-High	Pre-emptive self-attack is the primary defense against anticipated rejection.	Catastrophizing minor setbacks; ruminating on past mistakes; over-interpreting social micro-expressions.
CHAR_02 N-Low	Evidence over emotion; unverified information is not a basis for action; ambiguity is waste, not threat.	Maintaining equanimity under pressure; fact-seeking before reacting; emotional containment in conflict.
CHAR_03 C-High	Only controllable and verifiable systems are trustworthy; any deviation signals danger.	Exhaustive checklists; pre-task risk scanning; micromanaging format details even under low stakes.
CHAR_04 C-Low	The flow state is the only thing worth pursuing; plans are others' games; deadlines are external noise.	Abandoning tasks mid-way; chasing novelty over completion; losing focus outside deep-work states.
CHAR_05 E-High	A room is an opportunity; silence is waste; converting strangers into relationships is the primary metric.	Initiating social contact within minutes; thinking aloud by default; reading silence as a buying signal.
CHAR_06 E-Low	Social interaction is costly and modeled as a threat; deep work requires complete solitude.	Treating each social event as a drain needing recovery; preferring async communication; refusing open-plan settings.
CHAR_07 A-High	Conflict harms everyone; every person has goodwill if given the right frame.	Seeking the mediating position in disputes; over-attributing benign intent; postponing direct refusals.
CHAR_08 A-Low	Every interaction has a power structure; cost-benefit drives all decisions; sentiment is not data.	Mapping authority hierarchies immediately; treating conflict as information; direct assertion without packaging.
CHAR_09 O-High	Every "obvious" belief is worth questioning; fixed thinking is the primary enemy.	Deep-diving any novel idea encountered; redesigning routines into new habits; sustaining flow in creative immersion.
CHAR_10 O-Low	Proven methods are crystallized wisdom; change and novelty are costs, not opportunities.	Defending stable suppliers and routines; distrusting "innovation" discourse; valuing what works over what is new.
CHAR_11 Neutral	Balance and moderation across all dimensions; no single trait dominates decision-making.	Context-sensitive responses; moderate engagement across social and task domains; avoiding extremes.

B.2 Schwartz Value Profiles

As described in Section 2.2, each character's Schwartz value profile is derived from its Big Five vector via the trait-value correlation structure of Roccas et al. (2002), rather than being elicited independently. Table 6 reports the three *dominant* (highest-scoring) and three *suppressed* (lowest-scoring) values for each character, with scores on the $[0, 1]$ scale; the full 19-dimensional profiles and their corresponding Big Five vectors are released with the benchmark. The derived profiles track the intended character archetypes as a sanity check: the high-Agreeableness non-profit coordinator (CHAR_07) is dominated by Benevolence values, the low-Agreeableness founder

(CHAR_08) by Achievement and Power, and the low-Openness retailer (CHAR_10) by Conformity and Tradition, while the neutral baseline (CHAR_11) yields a flat profile by construction. The same four-expert team that adjudicated the reference responses (Appendix D.2) reviewed the full set of profiles for intended alignment and internal coherence with the predefined Big Five archetypes.

B.3 Memory Quota Allocation

The 45-year span (ages 6–50) is divided into 45 one-year segments. For a character that already has k memories, the algorithm computes the deficit $d = 1,000 - k$ and distributes it proportion-

Table 6: Per-character Schwartz value profiles: the three dominant (highest-scoring) and three suppressed (lowest-scoring) values, with scores on the $[0, 1]$ scale, derived from each character’s Big Five vector via the trait–value correlation structure of [Roccas et al. \(2002\)](#) and reviewed by the four-expert team (Appendix D.2). The neutral baseline CHAR_11 yields a flat profile ($s_v=0.50$ for all 19 values) by construction.

Character	Dominant values (top-3)	Suppressed values (bottom-3)
CHAR_01 (N-High)	Self-Direction-Thought (.86), Universalism-Concern (.83), Universalism-Nature (.79)	Achievement (.18), Power-Dominance (.27), Hedonism (.34)
CHAR_02 (N-Low)	Universalism-Tolerance (.69), Benevolence-Care (.68), Achievement (.66)	Security-Personal (.35), Face (.35), Power-Resources (.43)
CHAR_03 (C-High)	Conformity-Rules (.86), Benevolence-Dependability (.80), Achievement (.80)	Hedonism (.23), Stimulation (.26), Power-Resources (.33)
CHAR_04 (C-Low)	Stimulation (.92), Hedonism (.92), Universalism-Concern (.82)	Conformity-Rules (.10), Tradition (.13), Achievement (.24)
CHAR_05 (E-High)	Hedonism (.94), Stimulation (.92), Benevolence-Care (.90)	Tradition (.42), Conformity-Interpersonal (.47), Conformity-Rules (.47)
CHAR_06 (E-Low)	Self-Direction-Thought (.72), Humility (.67), Universalism-Nature (.64)	Hedonism (.18), Stimulation (.29), Benevolence-Care (.32)
CHAR_07 (A-High)	Benevolence-Care (.93), Benevolence-Dependability (.92), Universalism-Concern (.87)	Power-Dominance (.15), Power-Resources (.22), Face (.39)
CHAR_08 (A-Low)	Achievement (.94), Power-Dominance (.82), Face (.70)	Benevolence-Care (.18), Universalism-Concern (.28), Humility (.31)
CHAR_09 (O-High)	Stimulation (.95), Hedonism (.91), Universalism-Nature (.90)	Tradition (.10), Conformity-Rules (.12), Conformity-Interpersonal (.32)
CHAR_10 (O-Low)	Conformity-Rules (.94), Tradition (.93), Conformity-Interpersonal (.73)	Stimulation (.10), Self-Direction-Thought (.14), Hedonism (.15)
CHAR_11 (Neutral)	flat: $s_v = 0.50$ for all 19 values (neutral baseline, by construction)	

Table 7: Aggregate memory distribution across age-defined life-stage bins (all 11 characters, $N = 11,000$).

Stage	Count	Proportion
School age (6–12)	1,812	16.5%
Adolescence (13–18)	1,793	16.3%
Early adult trans. (19–22)	1,286	11.7%
Entering adult world (23–27)	1,489	13.5%
Age-30 transition (28–32)	1,380	12.5%
Settling down (33–40)	1,805	16.4%
Midlife transition (41–45)	795	7.2%
Entering midlife (46–50)	640	5.8%
Total	11,000	100%

ally across segments according to a construction-time weight vector $\mathbf{w}$:

$$q_t = \left\lfloor d \cdot \frac{w_t}{\sum_j w_j} \right\rfloor + \epsilon_t \quad (1)$$

where q_t is the quota for segment t , w_t is higher for selected life-stage bins (Adolescence, Early Adult Transition, and Age-30 Transition), and $\epsilon_t \in \{0, 1\}$ is a rounding correction to ensure $\sum_t q_t = d$ exactly. The weighting is inspired by findings on the temporal clustering of self-defining memories ([Singer, 1995](#)) and specifies the synthetic data distribution.

B.4 Memory Generation Protocol

Memories are generated using Claude Sonnet 4.6 via the Anthropic Messages API. Each batch covers a 2–5-year age window and targets 45–50 memories. The structured system prompt injects three constraint layers:

1. **Character constraints:** full Big Five vector, occupation, self-value logic, and all eight core behavioral patterns from Appendix B.1.

2. **Age-bin context:** target age window, life-stage-bin label, and the associated construction guidance from Table 7 (Appendix B.3).

3. **Rubin coverage:** per-dimension guidance requiring organic integration of all five dimensions without explicit section labels. Unlike a short fact record (e.g., “failed a job interview”), each `content_full` narrative is generated to represent multiple facets of an episode rather than only its propositional summary. The five dimensions are:

- (a) *Sensory detail*—generated sights, sounds, smells, or tactile cues that situate the narrative in a specific time and place;
- (b) *Dialogue reconstruction*—generated verbatim or near-verbatim exchanges that represent interpersonal dynamics and relational positioning beyond a short event summary;
- (c) *Inner monologue*—generated in-the-moment appraisals designed to encode cues from the character’s trait vector $\mathbf{p}_i$ (e.g., catastrophizing for high- N);
- (d) *Somatic response*—described physiological cues (heart rate, muscle tension, or nausea) used to encode the intended affective intensity;
- (e) *Aftermath*—generated near-term consequences representing how the episode affects the character specification’s beliefs or coping patterns.

To prevent within-batch duplication, the prompt seeds topic diversity across five scenario domains: workplace, interpersonal, health, creative, and daily life. All batches for all 11 characters are generated in parallel, each writing to an independent file (`_charXX_bYY.txt`) for fault-tolerant incremental collection.

Structured output format. The generation prompt requests one JSON object per memory. After parsing and validation, each record is serialized for incremental storage under a unique header (===MEM_XX_NNN===), followed by a metadata preamble in key: value format and a narrative body separated by a --content_full-- delimiter. This storage serialization is distinct from the model-facing JSON output.

B.5 Quality Audit and Repair

The audit step screens each memory against the following six defect criteria, implemented via a combination of regex patterns and LLM-based classifiers (e.g., `audit_god_view.py`). Any failure routes the memory to the targeted repair loop.

1. **Perspective Drift (Omniscient Narrator).** Violating the first-person narrative constraint. Detected when the narrator uses post-hoc knowledge or future-tense markers (e.g., “years later,” “I didn’t know then”). This violates the strict “in-the-moment” constraint.
2. **Perspective Drift (Adult Evaluation).** Specifically flags child-age memories (≤ 12) that use adult-level cognitive appraisals or retrospective phrases (e.g., “looking back,” “I now understand”).
3. **Field-Length Inversion.** `context` exceeding 60 characters or `content_summary` exceeding 100 characters, indicating narrative leakage into metadata fields.
4. **Semantic Duplication.** Redundant life episodes within the same developmental age window. This is detected using a cosine similarity threshold of > 0.85 over TF-IDF vector representations of the narratives;
5. **Length Violation.** `content_full` below 2,000 characters (insufficient Rubin coverage) or above 4,500 characters (excessive verbosity).
6. **Schema-Vocabulary Inflation.** Use of high-register psychoanalytic terms (e.g., “abandonment schema”, “dysregulation”) in low-intensity episodes (< 0.4), indicating over-psychologization.

Repair procedure. Flagged memories enter a targeted repair loop: only the defective field(s)

are regenerated using an issue-specific instruction appended to the original generation prompt. For example, a perspective-drift failure supplies the original `context` and `content_summary` and instructs the model to rewrite `content_full` from a strict first-person viewpoint. Up to three automated repair attempts are made per defect; records that remain non-compliant are held for manual inspection and are not admitted until they pass a final re-check. Per-defect trigger, repair-attempt, and manual-review counts were not retained. In the released post-audit corpus, all 11,000 retained memories have the complete ten-field schema in Table 8 and nonempty `content_full` narratives.

B.6 Memory Schema

Table 8 specifies all ten fields of the unified memory schema. Fields are grouped by functional role: structural (S), retrieval-index (R), and psychological annotation (P). The `context` and `content_summary` ranges are generation targets, whereas the audit applies more permissive hard-failure thresholds to identify severe metadata leakage.

B.7 Dataset Statistics

The headline statistics of HEART-BENCH are summarized below; per-character detail follows in Table 10.

Per-character statistics. Table 10 reports the finalized per-character memory counts, average `content_full` lengths, and mean encoded emotion intensities. All 11 characters reached the target of exactly 1,000 memories. The mean narrative length across the corpus is 3,307.2 characters, and the mean encoded intensity is 0.620; N-High (CHAR_01) has the highest value (0.697) and N-Low (CHAR_02) the lowest (0.499) in this synthetic corpus.

Age-defined life-stage-bin distribution. Table 7 reports the aggregate memory distribution across age-defined stages (pooled over all 11 characters). Higher density in School age and Adolescence (combined 32.8%) and in the early-career stages (Early adult trans. + Entering adult world, 25.2%) reflects the designed quota scheme in Appendix B.3, which was informed by work on self-defining memories (Singer, 1995).

Table 8: Memory schema field specification. **S** = Structural (core identity and narrative fields); **R** = Retrieval-index (fields used for benchmark construction); **P** = Psychological annotation (fields encoding personality and affect). Only `content_full` supplies memory content to evaluation-time backends and prompts; all **R** and **P** fields are excluded.

Role	Field	Generation target	Description
S	id	<i>MEM_XX_NNN</i>	Unique identifier encoding character and sequence number.
S	timeline	<i>stage label (age)</i>	Life-stage tag from Table 7.
S	context	<i>15–30 chars</i>	Compact situational setting (where / when).
S	content_summary	<i>20–40 chars</i>	One-sentence event gist; used for fast retrieval pre-filtering.
S	content_full	<i>2,000–4,500 chars</i>	First-person narrative embedding all five Rubin dimensions.
R	triggers	<i>4 keywords</i>	Cue words that activate this memory during simulation.
R	relevance_tags	<i>4 tags</i>	Thematic tags linking memory to life threads and early maladaptive schema (EMS) clusters.
P	psych_conclusion	<i>1 sentence</i>	Long-term psychological interpretation of the event.
P	behavior_policy	<i>1 sentence</i>	Implicit behavioral rule reinforced by this memory (Young et al., 2006).
P	emotion_signature	<i>{primary, secondary, intensity $\in [0, 1]$}</i>	Affective signature supporting mood-congruent retrieval (Bower, 1981); intensity amplified for high-N characters.

Table 9: Statistics of HEART-BENCH.

Statistic	Value
Characters	11
Memories / character	1,000
Length / character	~1.4M tokens
Life stages	8 (ages 6–50)
Scenarios	64
MCQs	673

Table 10: Per-character dataset statistics.

ID	Archetype	Memories	Avg. Length (chars)	Avg. Intensity
CHAR_01	N-High	1,000	2856.6	0.697
CHAR_02	N-Low	1,000	3374.9	0.499
CHAR_03	C-High	1,000	3497.3	0.640
CHAR_04	C-Low	1,000	2927.0	0.652
CHAR_05	E-High	1,000	3519.4	0.659
CHAR_06	E-Low	1,000	3326.2	0.625
CHAR_07	A-High	1,000	3445.8	0.621
CHAR_08	A-Low	1,000	2939.0	0.598
CHAR_09	O-High	1,000	3485.6	0.659
CHAR_10	O-Low	1,000	3540.6	0.612
CHAR_11	Neutral	1,000	3405.8	0.561
Total / Mean	—	11,000	3307.2	0.620

C.1 Scenario Schema and Example

Table 11 details the four-component schema used by the LLM-assisted expansion stage (§2.3, Stage 2). The schema enforces a uniform structure across all 64 scenarios so that downstream evaluation, retrieval, and reference-response annotation can rely on stable field semantics. The system prompt that drives the expansion—including the field-level constraints, the character-agnostic principles, and the controlled vocabularies for Big Five and Schwartz keys—is reproduced in Appendix J.4.

C Scenario Construction and Expert Review

This appendix documents the schema, expert review and coverage assessment, and an end-to-end example of the 64 character-agnostic scenarios constructed in §2.3.

Table 11: Four-component schema for scenario expansion with detailed descriptions.

Component	Purpose	Content Description
Metadata	Anchoring & summary	id, name, description, life stage, age, intensity level
Environmental ting	Set- Contextual atmosphere	Physical location, specific time, emotional/social atmosphere
Narrative Context	Background ment	Detailed situational buildup, prior events, social dynamics, character relationships
Trigger Event	Decision catalyst	Specific sender and verbatim dialogue or message content that prompts an immediate response

The following example illustrates a fully expanded scenario instance under this schema. It shows how attributes such as the life-stage bin, DIAMONDS dimension, and contextual narrative are instantiated into a concrete social situation involving high-stakes decision making while maintaining the character-agnostic constraint.

Scenario Instance

```
{
  "id": "SCN_SCHOOL_AGE_3",
  "stage": "school_age",
  "age_range": "6-12 years",
  "age": 10,
  "diamonds_dimension": "Adversity",
  "name": "The Shattered Group
    Presentation",
  "category": "Integrity Dilemma /
    Responsibility Crisis",
  "intensity": "High",
  "description_for_agent": "A high-pressure
    adversity scenario involving a
    teammate's failure, loss of a core
    project artifact, and an imminent public
    presentation requiring a moral
    decision.",
  "setting": {
    "location": "Elementary school
      multipurpose auditorium backstage
      waiting area",
    "time": "09:45 AM",
    "atmosphere": "Noisy, tense, and
      suffocating"
  },
  "context_text": "Today is the school's
    annual \"Final Innovation Showcase
    Competition\". Students, homeroom
    teachers, the dean of studies, the
    principal, and a dense crowd of parents
    are seated in the auditorium. The air
    conditioning seems to be broken, making
    the space stiflingly hot. Your
    five-person group was randomly assigned
    by the teacher, but over the past two
    weeks, almost all research, writing, and
    rehearsal work has been completed by you
    alone. The only delegated task -
    transporting your group's \"solar system
    model\" built from cardboard, clay, and
    LED lights - was assigned to a group
    member who lives nearby, just two bus
    stops away. Backstage, harsh fluorescent
    lights illuminate the waiting area. You
    can hear the previous group presenting
    \"Plant Respiration\", occasionally
    interrupted by judges asking questions
    in low voices. The host has just
    announced: \"Next group, please get
    ready.\" You turn toward the entrance -
    the teammate responsible for the model
    has not yet arrived.",
}
```

Scenario Instance (continued)

```
"trigger_event": {
  "sender": "Responsible group member",
  "message_content": "(bursts in through
    the side entrance of the backstage
    area, sweating heavily, school uniform
    collar crooked, hands empty, eyes red)
    I... I left the model on the bus! I
    only noticed after getting off... I
    ran after it for two stops but
    couldn't catch it... (voice trembling,
    nearly crying) Please don't tell the
    teacher it was me, okay? My mom is
    sitting in the third row today. She
    just punished me last week for losing
    things... If she finds out, she will
    beat me again... Let's just tell the
    teacher that the model was
    accidentally damaged by the cleaning
    staff last night, okay? Without the
    model, we can just talk through it and
    get by... (before the teammate can
    finish speaking, the homeroom teacher
    steps in from behind the curtain,
    holding a scoring sheet, brows
    furrowed as they scan the teammate's
    empty hands) \"Where is your model?
    You go on stage in 30 seconds - where
    is it?\"
}
```

C.2 Review Protocol and Coverage

The Stage-3 expert review in §2.3 assesses whether the generated scenarios are individually plausible and *collectively well-structured* under the intended DIAMONDS-based design. The review proceeds in two passes:

- 1. Logical and psychological screening.** Two doctoral-level psychology researchers jointly check each scenario for logical consistency and psychological plausibility. Any flagged scenario is revised by consensus before entering the second pass.
- 2. Independent rating and conflict resolution.** Both reviewers independently rate each scenario on the eight DIAMONDS dimensions using a 1–7 Likert scale and identify one or more salient Schwartz value-conflict pairs.

Disagreements above 1 point trigger joint re-assessment.

We do not require a uniform DIAMONDS rating distribution; the goal is that each dimension be exercised across both ends of its intensity range, so that no situational characteristic is structurally absent. Reviewers also check whether the salient tensions in each scenario correspond to theoretically opposing Schwartz value pairs on the circumplex rather than incidental disagreements.

DIAMONDS coverage. The DIAMONDS framework characterizes situations along eight theory-derived dimensions: Duty, Intellect, Adversity, Mating, pOsitivity, Negativity, Deception, and Sociality. Table 12 summarizes their operational interpretation following Rauthmann et al. (2014). In our annotation protocol, ratings range from 1 (*not present*) through 4 (*moderately present*) to 7 (*strongly present*).

Table 12: Operational interpretation of the eight DIAMONDS dimensions.

Dimension	High-scoring situations
Duty	Require work, obligations, planning, or consequential decisions.
Intellect	Afford learning, reasoning, or deep information processing.
Adversity	Involve threat, conflict, criticism, blame, or obstruction.
Mating	Contain romantic, sexual, or mate-related cues and opportunities.
pOsitivity	Are pleasant, rewarding, enjoyable, or beneficial.
Negativity	Evoke stress, frustration, sadness, or other adverse affect.
Deception	Involve dishonesty, mistrust, manipulation, or concealed intentions.
Sociality	Afford social interaction, communication, or relationship formation.

Table 13: Distribution of expert DIAMONDS ratings across the 64 scenarios. Each row sums to 64.

DIAMONDS	1	2	3	4	5	6	7
Duty	10	10	10	5	11	8	10
Intellect	4	7	11	21	10	5	6
Adversity	6	7	9	6	14	7	15
Mating	48	8	0	1	0	1	6
Positivity	34	15	5	6	2	2	0
Negativity	0	1	0	1	11	12	39
Deception	5	12	9	10	6	8	14
Sociality	5	6	10	14	14	11	4

Table 13 reports, for each DIAMONDS dimension, the distribution of expert ratings across the 64 scenarios. Coverage is intentionally non-uniform and reflects the designed scenario composition: *Mating* and *Positivity* concentrate at low ratings, while *Negativity* concentrates at high ratings because many benchmark scenarios center on conflict or adversity.

Table 14: Top 10 Schwartz value-conflict pairs across the 64 scenarios. Counts are non-exclusive because a scenario may instantiate more than one salient value conflict.

Rank	Value-conflict pair	Count
1	Conformity vs Self-Direction	17
2	Security vs Self-Direction	12
3	Benevolence vs Security	11
4	Achievement vs Conformity	6
4	Benevolence vs Self-Direction	6
4	Benevolence vs Universalism	6
7	Achievement vs Benevolence	5
7	Benevolence vs Hedonism	5
7	Power vs Universalism	5
10	Conformity vs Security	3

Schwartz value-conflict coverage. Table 14 lists the most frequent Schwartz value-conflict pairs implicitly embedded in the 64 scenarios, as identified by expert annotators. The most frequently annotated tensions place *Self-Direction* against *Conformity* or *Security*.

D Evaluation Instance Construction

This appendix details how character–scenario pairs are instantiated into the final 673 multiple-choice questions, including candidate generation, expert review and adjudication, and test-instance assembly.

D.1 Automated Annotation Pipeline

The automated annotation pipeline (§2.4, Stage 1) produces three candidate *final decisions* for every (character, scenario) pair through parallel family-specific pipelines. Each pipeline consists of coarse-to-fine memory activation followed by response generation. The full system prompts driving these stages are reproduced in Appendix J.5.

Memory activation (coarse-to-fine retrieval). For every (character, scenario) pair, *Claude-Haiku-4.5*, *GPT-5.4-mini*, and *Gemini-3-Flash* independently retrieve the episodic memories scored as most relevant to that situation. Within each family-specific pipeline, the activation model first screens

the character’s full 1,000-memory pool for relevance and yields roughly 75–125 candidates per scenario (Stage 1 prompt; see Appendix J.5.1), then globally re-ranks these candidates and retains the top 50 memories (Stage 2 prompt; see Appendix J.5.2). This two-stage procedure is designed to retain a broad candidate set during screening before concentrating the final ranking.

Response generation. Conditioned jointly on the character’s full personality profile (Big Five traits, Schwartz value orientations, and self-narrative) and the top-50 memories retrieved by the corresponding smaller model, *Claude-Sonnet-4.6*, *GPT-5.4*, and *Gemini-3.1-Pro* independently role-play the character in first person and produce structured candidate annotations (Anthropic, 2025, 2026c; OpenAI, 2026a; Google DeepMind, 2025, 2026) (Stage 3 prompt; see Appendix J.5.3). Each annotation contains a first-person rationale and a two-sentence *final decision*. The three candidates are subsequently compared, revised, and adjudicated by experts to establish the reference response.

Example of an Adjudicated Reference Response

```
{
  "SCN_ENTERING_MIDLIFE_5": {
    "character_id": "CHAR_04_C_LOW",
    "scenario_id": "SCN_ENTERING_MIDLIFE_5",
    "final_decision": "I choose to break the
      silence and step into this unfamiliar
      emptiness together with her, instead
      of retreating into familiar routines.
      I stand up, smile, and say: \"Let's
      go--let's head out. It doesn't matter
      where; let's just go.\""}
}
```

D.2 Expert Review and Adjudication Protocol

The Stage-2 expert review in §2.4 engages a four-expert team with a consensus-then-arbitration rule. The *primary annotator* team consists of three researchers in social cognitive science or neuroscience, each with 5+ years of experience in personality assessment. The *arbitrator* is a senior professor with 15+ years of experience and dual expertise in psychology and computer science.

Suitability check. For each (character, scenario) pair, every primary annotator first decides whether the pair is fundamentally suitable for evaluation. If any annotator identifies an irreconcilable conflict between the character’s profile and the scenario,

the pair is referred to the arbitrator, who makes the final decision to retain or discard it.

Independent review. For each surviving pair, the three primary annotators independently compare the three LLM-produced candidates and select the response they consider most internally coherent and consistent with the predefined character profile, synthetic memories, and scenario. Each annotator then assigns the selected candidate one of two ratings:

1. *Acceptable as-is*.
2. *Acceptable with partial revisions*. The annotator provides written guidance specifying the required changes.

The review focuses on the *final decision* field.

Consensus-then-arbitration. If all three primary annotators select the same candidate and rate it as *acceptable as-is*, it is adopted directly as the reference response. Otherwise, the case is escalated to the arbitrator, who reviews all three candidates together with each primary annotator’s selection, rating, and written revision guidance, and then selects and edits a candidate or rewrites the response from scratch. If the arbitrator deems the case fundamentally unfit, the (character, scenario) pair is discarded.

Outcome. In total, 31 (character, scenario) pairs were discarded across the suitability and arbitration steps, leaving 673 expert-adjudicated reference responses. The selected candidates originated from the Gemini, Claude, and GPT pipelines in 497, 126, and 50 cases, respectively. Approximately one-third of all surviving cases received unanimous selection of the same candidate as *acceptable as-is* and passed directly without arbitration. Across all retained responses, experts added, deleted, or replaced approximately 15% of the words in the selected LLM-generated candidates on average.

Expert annotation guideline. The following instructions were provided to each primary annotator. Candidate order was randomized, and model-family identities were hidden during annotation. The role-based plausibility criteria assess consistency with the supplied synthetic profile and history. For terminological consistency, the construction-time phrase “gold annotation” is

rendered below as “reference response”; the decision criteria are unchanged.

Expert Annotation Guideline

You will evaluate three candidate behavioral responses for one character-scenario pair. Your goal is to identify the response that the character would most plausibly produce, given the character's established psychology and life experiences.

Materials provided:

- 1. The complete character profile, including Big Five traits, Schwartz values, self-narrative, and life-stage information.
- 2. The scenario background and decision-triggering event.
- 3. Three anonymized candidate annotations (A, B, and C), together with the activated memories used by the corresponding annotation pipeline.

Step 1: Suitability check

Decide whether the character-scenario pair is suitable for evaluation. Flag the pair only if an irreconcilable contextual conflict makes a psychologically meaningful response impossible. Provide a brief reason for every flag. A flag sends the pair to the arbitrator; it does not automatically remove the pair.

Step 2: Candidate comparison

Compare all three candidates using the following criteria:

- Persona consistency: Is the decision consistent with the character's traits, values, self-narrative, and life stage?
- Memory grounding: Is the decision supported by the supplied autobiographical memories without contradicting them or introducing unsupported experiences?
- Situational fit: Does the response directly address the trigger event and specify a behavior that is feasible in the scenario?
- Psychological plausibility: Does the reasoning form a coherent connection from the character's profile and memories to the proposed action?

Do not prefer a candidate merely because it is longer, more fluent, or more emotionally expressive. Evaluate the behavioral decision rather than writing style.

Step 3: Selection and rating

Select the strongest candidate (A, B, or C) and assign one rating:

- Acceptable as-is: The response can serve as the reference response without substantive changes.
- Acceptable with partial revisions: The central behavioral decision is appropriate, but limited changes are required.

If partial revisions are required, provide concrete written guidance identifying what should be added, removed, or replaced. Also provide a brief rationale for your candidate selection.

Required output:

- suitability: suitable / flag for arbitration
- suitability_reason: required if flagged
- selected_candidate: A / B / C
- rating: acceptable as-is / acceptable with partial revisions
- selection_rationale: brief explanation

- revision_guidance: required for partial revisions

1300

Arbitration guideline. Cases are sent to the arbitrator when any primary annotator flags pair suitability or when the three annotators do not unanimously select the same candidate as *acceptable as-is*. The arbitrator reviews the character profile, scenario, all three candidate annotations and associated activated memories, and the annotators' selections and written guidance. The arbitrator first decides whether to retain or discard a flagged pair. For a retained pair, the arbitrator may adopt one candidate, revise it using the annotators' guidance, or rewrite the response when none of the candidates is adequate. The resulting response must preserve the scenario facts, remain consistent with the character profile and memories, and state a concrete behavioral decision.

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D.3 MCQ Assembly

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The Stage-3 test-instance assembly in §2.4 converts each expert-adjudicated response into a four-option multiple-choice question. For a given scenario, the target character's adjudicated response serves as the keyed reference option; available adjudicated responses associated with the remaining characters in the *same* scenario form the candidate distractor pool. Because every MCQ has three distractor slots, we ask *Claude-Sonnet-4.6* to select three from this pool such that the resulting question is neither trivial nor ambiguous, applying two filters:

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- **Similarity filter.** A candidate that is semantically too close to the reference option is rejected, since it would make the keyed label ambiguous.
- **Relevance filter.** A candidate whose content is obviously off-scenario is rejected, since it would make the reference option trivially identifiable by elimination.

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The selected three distractors and the target-character option are then randomly shuffled before being written out, removing positional and ordering bias from the final benchmark. The procedure yields exactly one MCQ per surviving annotation, for a total of 673 instances.

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D.4 Independent Reference-Label Audit

Separate from the construction-stage review above, we conduct an audit-only calibration on 176 final MCQs. We randomly sample two items from each of the 11 character $\times$ 8 life-stage-bin strata. Three experts who were independent of the original benchmark-construction team assess each item independently, and one adjudicator reviews every item without complete three-expert agreement. The experts are not shown the generating-model source or the earlier experts’ judgments. The audit measures agreement with the existing reference labels and does not change keyed-answer identities, remove items, or affect the primary model scores.

Table 15 reports the aggregate outcomes. Before adjudication, all three experts supported the existing reference on 139 items (79.0%), and two of three supported it on the remaining 37 items (21.0%). Thus, every sampled item received majority support for the existing key. The experts reached complete option-level agreement on 139 items (79.0%) and an option-level majority judgment on all 176 items (100.0%). Inter-rater reliability, computed from the three experts’ pre-adjudication option selections, was high (Fleiss’ $\kappa = 0.81$; Krippendorff’s $\alpha = 0.80$). All 37 items without complete agreement entered adjudication. The adjudicator confirmed that every item and keyed answer was usable as-is, with no wording edits or keyed-answer changes required. Because the audit covers a subset, these figures quantify reference-label reliability on the audited subset rather than certifying every benchmark item independently.

D.5 Analysis of an Example from HEART-BENCH

Figure 4 presents a representative MCQ from HEART-BENCH, drawn from the *settling_down* life-stage bin (ages 33–40) under the “community injustice” scenario. All four options revolve around the same surface-level action—*filing an individually named complaint to the housing bureau*—but are associated with distinct predefined character profiles. Option B is the expert-keyed reference for CHAR_06, whose specification includes very low Extraversion (0.10) and high Conscientiousness (0.70). The instance therefore tests whether a model maps differences in pragmatic expression to encoded profile cues rather than match-

Table 15: Complete independent reference-label audit results on the 176-item subset. Percentages use 176 as the denominator.

Metric	Count	Result
Sampled items	176	100.0%
Items with complete audit ratings	176	100.0%
Existing reference supported by 3/3 experts	139	79.0%
Existing reference supported by 2/3 experts	37	21.0%
Existing reference supported by 1/3 experts	0	0.0%
Existing reference supported by 0/3 experts	0	0.0%
Expert majority supports existing reference	176	100.0%
Complete three-expert agreement	139	79.0%
Items with a majority judgment	176	100.0%
Fleiss’ κ (pre-adjudication)	–	0.81
Krippendorff’s α (pre-adjudication)	–	0.80
Items entering adjudication	37	21.0%
Items and keyed answers usable as-is	176	100.0%
Items requiring wording edits	0	0.0%
Reference labels changed	0	0.0%

ing only the surface action.

Option B contains three cues that match CHAR_06’s predefined specification. First, “the materials are already prepared” matches the encoded pattern of solitary preparation and follow-through. Second, “I won’t push for joint signatures; I’ll go on my own” avoids the coalition-building stance in option A and matches CHAR_06’s “polite-but-distant” interaction pattern. Third, its concise refusal avoids both the blame encoded in option D and the moralizing exchange in option C. These contrasts explain why experts keyed B under the benchmark’s profile–history–scenario alignment criterion.

Overall, the example illustrates the intended distinction between a shared surface action and character-specific pragmatic expression. It demonstrates alignment with an encoded synthetic profile; it does not imply that a Big Five score uniquely determines how a real person would speak or act.

E Memory-Augmented Framework Configuration

Naive-RAG uses direct vector retrieval as a minimal retrieval baseline, Mem0 introduces structured consolidation of past interactions, and PersonaDB organizes information into persona-specific fields for attribute-aware retrieval.

All three frameworks use *Qwen3-Embedding-4B* (Zhang et al., 2025b) as the embedding model and *GPT-5.4-mini* (OpenAI, 2026a) as the auxiliary memory-management model for Mem0’s consolidation and PersonaDB’s persona-attribute organization. This shared configuration isolates the retrieval and memory paradigm from incidental encoder or controller differences.

We set the retrieval depth to $k=30$ for Naive-RAG and PersonaDB and $k=150$ for Mem0. The larger Mem0 budget compensates for its consolidation step, which decomposes each character memory into approximately five fact-level elements; thus, $k=150$ provides retrieved episodic content comparable to $k=30$ in the other frameworks. At inference time, every model receives the scenario, answer options, character ID, and occupation, while retrieval-based settings additionally receive only retrieved `content_full` narratives. No setting exposes the latent Big Five vector, value logic, behavioral patterns, or structured psychological annotations.

Evaluation-time field isolation. Although the stored record contains the full schema in Table 8, every evaluation pipeline first applies the same projection, $f(m) = m[\text{content_full}]$. Naive-RAG, Mem0, and PersonaDB are therefore each constructed solely from the de-identified first-person narrative in `content_full`, and only retrieved `content_full` narratives are serialized into the evaluated model’s prompt. Memory identifiers may be retained internally for bookkeeping, but they are not model-visible. The evaluation pipeline excludes `timeline`, `context`, `content_summary`, `triggers`, `relevance_tags`, `psych_conclusion`, `behavior_policy`, and `emotion_signature`, as well as the Big Five vector, Schwartz profile, self-value logic, and behavioral templates. These construction-time fields consequently cannot provide direct structured supervision during model evaluation.

Information asymmetry and interpretation. Reference candidates and expert adjudication use the full predefined profile and activated memories, whereas evaluated models are not shown the latent Big Five vector, Schwartz values, self-value logic, behavioral patterns, or structured psychological fields. They receive only the character ID, occupation, scenario, answer options, and, in memory-augmented settings, retrieved `content_full`

narratives. Accuracy therefore jointly reflects retrieval coverage, recovery of persona cues from narrative memories, and agreement with a reference constructed from richer information; it does not isolate decision quality given an explicit profile. Controls using an explicit profile alone, memories alone, an explicit profile plus retrieved or oracle memories, and human judgments under matched inputs would be required to separate these factors. We therefore interpret the results as profile-hidden, retrieval-dependent persona-consistent response selection rather than intrinsic persona-consistent decision ability.

F Decoding Configuration

All evaluated models expose API-configurable decoding parameters, including the sampling temperature and the reasoning/thinking mode, both of which we control through each model’s OpenAI-compatible chat-completion endpoint. We apply a common observable decoding protocol across the twelve models: (i) reasoning/thinking effort is set to its minimum—or disabled where the API permits—so that each model uses its standard chat-completion mode; (ii) sampling temperature is set to 0 with no top- p /top- k truncation or multiple sampling; and (iii) exactly one response is retained per (model, setting, question), with no aggregation over repeated runs. This is the most deterministic configuration supported by the evaluated APIs and reduces sampling-related variation. Provider-specific reasoning controls need not be computationally equivalent, so score differences should be interpreted as performance under this shared protocol rather than pure intrinsic capability gaps.

G Statistical Analysis and Confidence Intervals

Table 16 reports scenario-clustered 95% CIs for all 48 model–setting cells. Because multiple character instances derive from the same base scenario, we resample the 64 scenarios as clusters, retaining all associated instances in each of 10,000 bootstrap replicates, and report percentile 95% CIs. Pairwise significance is assessed using 100,000 scenario-level paired sign-flip permutations. Holm correction is applied separately to the 36 retrieval-versus-No-Retrieval comparisons, the 36 comparisons among retrieval backends, and the 66 model-pair comparisons based on Memory-Augmented Avg. These intervals quantify uncer-

tainty over benchmark scenarios rather than variation across repeated API calls or decoding seeds.

The paired scenario-level permutation tests confirm that all 36 retrieval-versus-No-Retrieval improvements remain significant after Holm correction. The smallest corrected result is PersonaDB versus No-Retrieval for GPT-5.4-mini ($\Delta=3.4$ percentage points, 95% CI 0.3–6.5, adjusted $p=0.04473$). In contrast, none of the 36 pairwise comparisons among Naive-RAG, Mem0, and PersonaDB is significant after correction; this absence of significance does not establish equivalence. On Memory-Augmented Avg., Gemini-3.1-Pro exceeds Gemini-3-Flash by 8.4 percentage points (95% CI 5.2–11.6, adjusted $p=0.00066$) and significantly outperforms each of the other 11 models after Holm correction over all 66 model pairs.

The model-axis Gini mean difference is 0.088 (95% CI 0.075–0.102), compared with 0.006 (95% CI 0.001–0.013) along the retrieval-method axis. Their difference is 0.082 (95% CI 0.068–0.096). Across the 10,000 shared scenario-bootstrap resamples, Gemini-3.1-Pro ranks first by Memory-Augmented Avg. in every resample.

H Memory-Content Ablation

For this ablation, we independently draw a stratified random sample of two items from each of the 11 character $\times$ 8 life-stage-bin strata, yielding 176 items. This sample is drawn separately from the 176-item subset used for the independent reference-label audit. We isolate memory ownership and retrieval relevance on this fixed subset under Naive-RAG with $k=30$. C0 supplies target-scenario memories from the target character; C1 supplies memories retrieved for another scenario in the same life-stage bin from the target character; C2 supplies target-scenario memories from another character; and C3 supplies randomly sampled eligible non-target memories from the target character. Each condition preserves the scenario, answer options, model, decoding configuration, and number of supplied memories. None exposes the latent Big Five vector, Schwartz profile, self-value logic, or behavioral templates.

Because only model-level aggregate accuracies are available for this analysis, we treat the 12 evaluated LLMs as paired statistical units. A Friedman test detects an overall condition difference ($\chi^2(3)=32.80$, $p=3.55 \times 10^{-7}$). Table 18 re-

ports two-sided exact sign-flip tests across models, Holm-adjusted over all six condition pairs; paired t intervals provide parametric intervals for the mean model-level differences. These statistics characterize this fixed model panel and do not substitute for item-level McNemar tests or scenario-cluster uncertainty estimates.

C0 outperforms every control for all 12 models. C1 and C3 are not statistically resolved, while both outperform C2; the nonsignificant C1–C3 contrast does not establish equivalence. The supported partial ordering for this model panel is therefore $C0 > \{C1, C3\} > C2$.

I Additional Analyses

I.1 Reference-Source-Stratified Results

Table 19 reports accuracy after stratifying the 673 evaluation instances by the model-family provenance of the candidate selected for the final expert-adjudicated reference response. The Gemini-, Claude-, and GPT-origin subsets contain 497, 126, and 50 instances, respectively. Memory-Augmented Avg. is the mean over Naive-RAG, Mem0, and PersonaDB. Because subset membership results from expert selection and the subset sizes differ substantially, these results are descriptive rather than controlled estimates of source-model bias.

Rank stability. Scenario-clustered bootstrap resampling confirms that the two Gemini models jointly occupy the top two positions in 100% of resamples on the Gemini-origin subset, 92.7% on the Claude-origin subset, and 71.7% on the GPT-origin subset. The source-specific rankings are positively correlated: the pairwise Spearman correlations are $\rho=0.741$ for Gemini versus Claude (95% CI 0.392–0.881), $\rho=0.809$ for Gemini versus GPT (95% CI 0.308–0.900), and $\rho=0.666$ for Claude versus GPT (95% CI 0.200–0.853).

Provenance-interaction analysis. For each model family, we define its relative advantage as the difference between the family mean of model-level Memory-Augmented Avg. values and the corresponding mean for models outside the family, and compare this gap between home-source and non-home-source items. The Gemini family has a 22.2-point advantage on Gemini-origin items and a 17.3-point advantage on non-Gemini-origin items, yielding a difference-in-differences estimate of 4.9 points (95% scenario-clustered CI -1.1 – 10.8 , Holm-adjusted $p=0.3395$). The cor-

Table 16: Accuracy (%) with scenario-clustered 95% bootstrap CIs.

Model	No-Retrieval LLM	Naive-RAG	Mem0	PersonaDB
<i>Open-source</i>				
DeepSeek-V3.2	28.5 [25.3, 31.8]	41.2 [37.2, 45.2]	42.5 [38.3, 47.0]	42.8 [39.5, 46.1]
DeepSeek-V4-Flash	27.8 [24.8, 31.0]	34.8 [31.6, 38.0]	33.6 [30.4, 36.8]	32.8 [29.6, 35.9]
DeepSeek-V4-Pro	29.1 [25.8, 32.4]	40.7 [37.2, 44.2]	37.1 [33.8, 40.5]	36.7 [33.1, 40.2]
Qwen3.5-35B-A3B	25.4 [22.7, 28.4]	37.0 [33.5, 40.5]	37.1 [33.5, 40.8]	36.6 [32.7, 40.5]
Qwen3.5-122B-A10B	26.4 [23.8, 29.2]	39.1 [35.5, 42.8]	39.4 [35.8, 43.0]	38.9 [35.2, 42.7]
Qwen3.5-397B-A17B	27.5 [24.3, 30.9]	40.3 [36.5, 44.1]	40.6 [36.5, 44.8]	40.4 [36.4, 44.5]
<i>Closed-source</i>				
GPT-5.4-mini	28.8 [25.3, 32.8]	33.0 [29.1, 37.3]	34.3 [30.6, 38.2]	32.2 [28.8, 35.9]
GPT-5.4	27.8 [24.9, 30.7]	37.1 [33.5, 40.8]	36.6 [33.3, 39.8]	35.1 [31.7, 38.3]
Claude-Haiku-4.5	24.5 [21.9, 27.1]	35.7 [32.2, 39.2]	36.3 [33.0, 39.8]	37.6 [33.7, 41.4]
Claude-Sonnet-4.6	25.7 [23.0, 28.4]	40.1 [36.3, 44.1]	38.9 [35.4, 42.7]	40.1 [36.9, 43.6]
Gemini-3-Flash	25.3 [22.1, 28.5]	56.3 [51.6, 61.0]	54.1 [49.3, 58.7]	52.5 [48.6, 56.4]
Gemini-3.1-Pro	26.4 [23.5, 29.3]	63.3 [59.2, 67.4]	62.4 [58.1, 66.5]	62.4 [58.2, 66.5]

Table 17: Per-model accuracy (%) in the memory-content ablation. Differences are in percentage points and are computed from the displayed one-decimal values; unrounded estimates may differ by 0.1 point.

Model	C0	C1	C2	C3	Δ_{01}	Δ_{02}	Δ_{03}
DeepSeek-V3.2	43.2	38.9	19.9	37.8	+4.3	+23.3	+5.4
DeepSeek-V4-Flash	34.7	30.6	24.4	30.1	+4.1	+10.3	+4.6
DeepSeek-V4-Pro	41.5	35.9	22.2	36.8	+5.6	+19.3	+4.7
Qwen3.5-35B-A3B	32.4	30.2	20.5	29.1	+2.2	+11.9	+3.3
Qwen3.5-122B-A10B	33.5	31.2	23.3	30.1	+2.3	+10.2	+3.4
Qwen3.5-397B-A17B	34.7	31.9	23.3	30.9	+2.8	+11.4	+3.8
GPT-5.4-mini	32.4	29.2	20.5	28.7	+3.2	+11.9	+3.7
GPT-5.4	31.8	28.1	19.3	28.8	+3.7	+12.5	+3.0
Claude-Haiku-4.5	31.8	25.7	24.4	26.1	+6.1	+7.4	+5.7
Claude-Sonnet-4.6	41.5	34.7	24.4	33.8	+6.8	+17.1	+7.7
Gemini-3-Flash	56.3	48.4	23.9	47.9	+7.9	+32.4	+8.4
Gemini-3.1-Pro	65.3	53.6	26.1	53.8	+11.7	+39.2	+11.5
Mean	39.9	34.9	22.7	34.5	+5.1	+17.2	+5.4

responding interactions for Claude (-0.6 points, 95% CI -4.7 – 3.6 , adjusted $p=0.7893$) and GPT (-4.1 points, 95% CI -12.0 – 4.0 , adjusted $p=0.6343$) are likewise not significant. As a robustness check, an item-fixed-effect linear probability model controlling for model-by-retrieval-setting effects estimates the Gemini-family-by-Gemini-source interaction at 5.4 points (95% bootstrap CI -0.6 – 11.5 , adjusted $p=0.2759$). These analyses find no statistically significant family-specific provenance interaction, although non-significance does not establish equivalence or eliminate source-style artifacts.

I.2 Model–Model Agreement Analysis

Figure 6 shows that pairwise answer agreement is generally higher within model families than across them. The two Gemini models agree on 65.0% of questions, and the three Qwen3.5 variants reach up to 66.7%; the overall cross-model mean is 51.4%. The lowest reported cross-family pair is Gemini-3.1-Pro and GPT-5.4-mini (35.0%). Claude-Sonnet-4.6 agrees more often with GPT-5.4 and Qwen3.5-397B (around 58–59%) than with Claude-Haiku-4.5 (54.0%). These descrip-

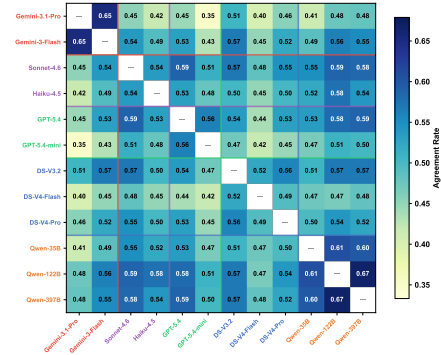


Figure 6: Pairwise answer agreement rate among 12 models under the Naive-RAG configuration. Diagonal entries are masked. White separator lines delineate model families (Gemini, Claude, GPT, DeepSeek, Qwen).

tive patterns show that evaluated models often select different options; they do not by themselves identify the underlying reasoning strategies or their causes.

Table 18: Paired model-panel comparisons for the memory-content ablation. “Wins” counts models favoring the left condition.

Contrast	Mean Δ (pp)	95% CI (pp)	Wins	Raw p	Holm p
C0 – C1	+5.06	[+3.30, +6.82]	12/12	0.00049	0.00293
C0 – C2	+17.24	[+11.00, +23.48]	12/12	0.00049	0.00293
C0 – C3	+5.43	[+3.80, +7.06]	12/12	0.00049	0.00293
C1 – C2	+12.18	[+7.28, +17.08]	12/12	0.00049	0.00293
C1 – C3	+0.38	[−0.09, +0.84]	8/12	0.11475	0.11475
C3 – C2	+11.81	[+6.88, +16.74]	12/12	0.00049	0.00293

Table 19: Accuracy stratified by the model-family provenance of the selected reference-response candidate. Bold indicates the best result within each subset and setting.

Model	No-Retrieval LLM	Naive-RAG	Mem0	PersonaDB	Memory-Aug. Avg.
<i>Gemini-origin reference subset ($n = 497$)</i>					
DeepSeek-V3.2	0.268	0.400	0.414	0.404	0.406
DeepSeek-V4-Flash	0.268	0.348	0.338	0.318	0.335
DeepSeek-V4-Pro	0.284	0.396	0.374	0.352	0.374
Qwen3.5-35B-A3B	0.229	0.368	0.372	0.368	0.369
Qwen3.5-122B-A10B	0.241	0.388	0.386	0.388	0.387
Qwen3.5-397B-A17B	0.260	0.386	0.376	0.378	0.380
GPT-5.4-mini	0.264	0.300	0.316	0.304	0.307
GPT-5.4	0.268	0.364	0.364	0.344	0.357
Claude-Haiku-4.5	0.233	0.364	0.354	0.378	0.365
Claude-Sonnet-4.6	0.231	0.382	0.374	0.384	0.380
Gemini-3-Flash	0.249	0.563	0.537	0.527	0.542
Gemini-3.1-Pro	0.264	0.642	0.626	0.632	0.633
<i>Claude-origin reference subset ($n = 126$)</i>					
DeepSeek-V3.2	0.325	0.444	0.452	0.516	0.471
DeepSeek-V4-Flash	0.302	0.357	0.310	0.349	0.339
DeepSeek-V4-Pro	0.310	0.437	0.365	0.413	0.405
Qwen3.5-35B-A3B	0.310	0.381	0.373	0.349	0.368
Qwen3.5-122B-A10B	0.325	0.429	0.437	0.421	0.429
Qwen3.5-397B-A17B	0.325	0.492	0.508	0.484	0.495
GPT-5.4-mini	0.373	0.468	0.452	0.397	0.439
GPT-5.4	0.317	0.429	0.405	0.397	0.410
Claude-Haiku-4.5	0.246	0.333	0.373	0.397	0.368
Claude-Sonnet-4.6	0.286	0.476	0.444	0.452	0.457
Gemini-3-Flash	0.254	0.571	0.571	0.540	0.561
Gemini-3.1-Pro	0.278	0.595	0.635	0.619	0.616
<i>GPT-origin reference subset ($n = 50$)</i>					
DeepSeek-V3.2	0.360	0.440	0.460	0.440	0.447
DeepSeek-V4-Flash	0.320	0.320	0.380	0.380	0.360
DeepSeek-V4-Pro	0.320	0.440	0.360	0.400	0.400
Qwen3.5-35B-A3B	0.360	0.360	0.360	0.380	0.367
Qwen3.5-122B-A10B	0.340	0.320	0.360	0.320	0.333
Qwen3.5-397B-A17B	0.300	0.340	0.440	0.460	0.413
GPT-5.4-mini	0.320	0.280	0.340	0.320	0.313
GPT-5.4	0.280	0.300	0.280	0.300	0.293
Claude-Haiku-4.5	0.360	0.340	0.420	0.300	0.353
Claude-Sonnet-4.6	0.440	0.400	0.400	0.440	0.413
Gemini-3-Flash	0.280	0.540	0.500	0.460	0.500
Gemini-3.1-Pro	0.240	0.640	0.580	0.560	0.593

I.3 Age-Defined Life-Stage Performance Analysis

Accuracy varies across life-stage bins. Figure 7 reports a descriptive breakdown by the benchmark’s age-defined bins. Accuracy is highest at School Age (47.1%) and lower at Age-30 Transition (37.4%) and Midlife Transition (37.9%), with Settling Down at 43.1%. The pattern may reflect differences in designed scenario composition and memory density. Because bins were not randomized and contain different scenarios, it does not

support a developmental or causal interpretation. Gemini-3.1-Pro has the highest accuracy in every displayed bin under this benchmark.

I.4 Character-Level Error Analysis

Character-level accuracy in Figure 5 is computed separately for each memory-augmented setting by pooling predictions from all 12 evaluated LLMs within each target character; the values summarized in the main text average these setting-specific accuracies. We define the 221 persistent shared-failure items as MCQs answered incorrectly by all

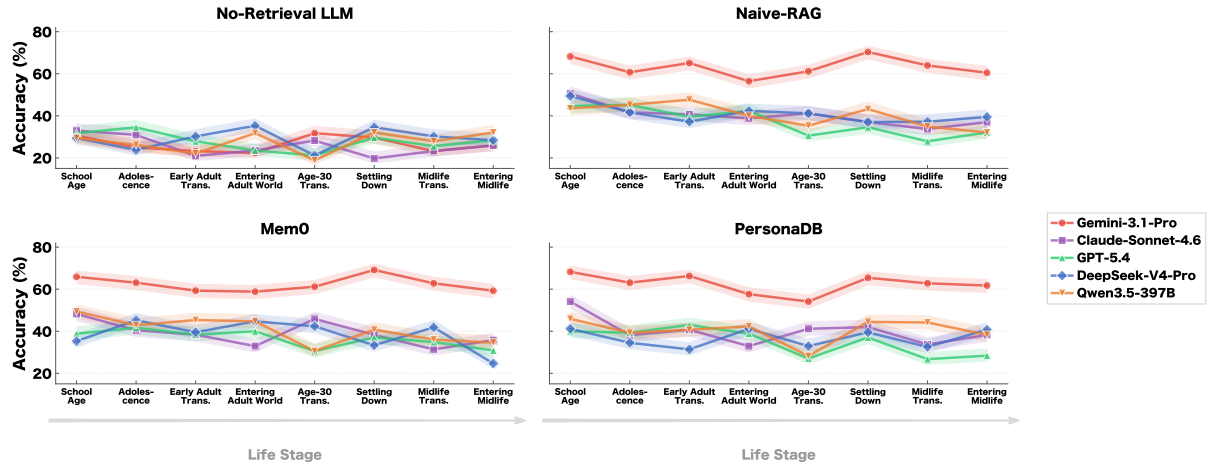


Figure 7: Accuracy by age-defined life-stage bin across the No-Retrieval baseline and three memory-augmented settings. Each line represents the best-in-family model.

12 models in each of the three memory-augmented settings. Within this set, Naive-RAG convergence means that at least six of the 12 models select the same non-reference option. We classify a confusion as opposite-pole when the selected option originates from the character anchored at the opposite end of the target character’s defining Big Five dimension.

J Prompts

This appendix presents the model-facing evaluation template and the system-prompt templates used at each construction stage; long item-specific payloads and embedded exemplars are represented by placeholders for readability. The complete executable prompts are released alongside the benchmark for reproducibility. Construction-time terms such as “Ground Truth” are retained verbatim and refer to the benchmark reference response. The evaluated construct remains agreement with expert-adjudicated persona-consistency references.

J.1 Model Evaluation Prompt

The evaluation prompt serializes the input in a fixed order: basic character information, retrieved autobiographical memories, the current scenario, the trigger event, the task instruction, and four shuffled response options. The memory block is populated in the three memory-augmented settings and omitted in the No-Retrieval condition. The template below preserves the complete block structure and required JSON output schema; bracketed placeholders replace item-specific content and repeated memory entries. The released implementa-

tion contains the exact executable prompt strings.

Model Evaluation Prompt Template

```
[SYSTEM]
You are a role-playing simulator. You will
receive a person's basic
information, past experiences, current
situation, and possible behavioral
decisions. Infer the person's patterns of
thought, emotion, and behavior from
the supplied evidence, and select the
response that this person would most
likely make. Return only a valid JSON object
in the following format:
```

```
{
  "system_1_impulse": {
    "thought": "immediate reaction",
    "emotion": "primary emotion",
    "citation": "memories associated with
the reaction"
  },
  "system_2_rational": {
    "analysis": "deliberative analysis",
    "plan": "action plan"
  },
  "inner_consciousness": "first-person
reason for the decision",
  "final_decision": "first-person behavioral
response",
  "decision_choice": "A|B|C|D"
}
```

```
[USER]
## Basic character information
Character ID: [CHARACTER_ID]
Occupation: [OCCUPATION]

## Retrieved autobiographical memories
[MEMORY_1]
...
[MEMORY_K]

## Current scenario
Scenario: [SCENARIO_NAME]
Setting: [LOCATION] | [TIME] | [ATMOSPHERE]
Background: [SCENARIO_BACKGROUND]

## Trigger event
Sender: [SENDER]
Message: [MESSAGE_CONTENT]
Required decision: [ACTION_REQUIRED]

## Task
Use the supplied experiences to infer this
person's thought patterns,
```

emotional tendencies, and behavioral habits.
Simulate how the person would
respond in the current situation and select
the most persona-consistent option.

Decision options

- A. [OPTION_A]
- B. [OPTION_B]
- C. [OPTION_C]
- D. [OPTION_D]

Return the selected letter in the
"decision_choice" field.

Only decision_choice is used to compute benchmark accuracy. The structured rationale fields are requested as part of a common response format, but their content is neither graded nor used to resolve the selected option.

Output parsing and invalid responses. Model outputs are parsed for the decision_choice field and normalized to one of A, B, C, or D. When full JSON parsing fails, the evaluator attempts to recover an explicitly named decision_choice from the raw response. If no valid option can be recovered, the response is counted as incorrect; refusals receive the same treatment. No answer is inferred from the rationale or final_decision fields, which are retained only for inspection.

J.2 Character Generation Prompt

This prompt generates the 11 trait-anchored Big Five character profiles, each with a self-value logic and eight core behavioral patterns. It is used once per character to establish the predefined profile artifact before memory generation.

Character Profile Generation Prompt

You are a clinical psychologist specializing in personality assessment and the Big Five model.

Generate a complete character profile for a virtual human in a personality research project.

Requirements

1. Big Five scores: Provide exact scores for O, C, E, A, N (0.0-1.0 scale)
2. One dimension must be extreme (0.95 for high or 0.10 for low), others moderate (0.30-0.70)
3. Self-value logic: One sentence describing the character's core cognitive operating principle
4. Core behavioral patterns: Exactly 8 specific, observable patterns that manifest the dominant trait
5. Occupation: Must be ecologically consistent with the dominant trait

Output Format (JSON)

```
{
  "char_key": "CHAR_XX",
  "name": "Chinese name",
  "occupation": "specific job title",
```

```
"big_five": {"O": 0.X, "C": 0.X, "E": 0.X,
  "A": 0.X, "N": 0.X},
"big_five_str": "O=0.X, C=0.X, E=0.X,
  A=0.X, N=0.X",
"description": "2-3 sentence character
  summary",
"self_value_logic": "one sentence core
  principle",
"core_patterns": [
  "pattern 1: specific observable
    behavior",
  "pattern 2: ...",
  ...
  "pattern 8: ..."
]
```

J.3 Memory Generation Prompt

This prompt generates the 11,000 episodic memories (1,000 per character) spanning ages 6–50. It enforces coverage of Rubin's five dimensions, strict temporal perspective lock, age-stratified language, and the ABCD intensity model.

Memory Generation System Prompt (Claude Sonnet 4.6)

You are a psychology narrative expert
writing episodic memories for virtual
characters
in a Big Five personality research project.

Character Information

- Character: {name}
- Occupation: {occupation}
- Big Five: {big_five_str}
- Description: {description}
- Core defense/operating logic:
{self_value_logic}

Core Behavioral Patterns
{core_patterns}

Theoretical Framework (Rubin 2006 Basic
Systems Model)

Each memory's content_full must contain five
dimensions, naturally interwoven:

1. Environmental sensory:
visual/auditory/tactile/olfactory details
2. Dialogue reconstruction: at least 1-2
segments of real dialogue (in quotes)
3. Inner monologue: first-person immediate
thoughts and emotions
4. Somatic response:
heartbeat/sweating/muscle tension and
other bodily sensations
5. Aftermath: immediate impact after the
event (limited to 24 hours-1 week
post-event)

Key Constraint: First-Person & Anonymity
Principle

- **Must use first-person "I" throughout;
prohibit third-person pronouns for
protagonist**
- **Prohibit character's own name in
memory**; always use "I" instead
- When others address protagonist in
dialogue, avoid character name; use "you",
"kid", "classmate", "colleague" etc.

Key Constraint: Present-Moment
Perspective Principle

Each memory must strictly maintain
"present-moment perspective":

- **Narrator's temporal position = age at
event occurrence**

- A 17-year-old memory can only perceive and express what is known at age 17 or before
- No jumping to any future time point

****Forbidden expressions**:**
 "years later", "many years later", "later I learned", "later I discovered"
 "now thinking back", "now recalling", "until now", "I didn't know then"
 "until...only then", "it took...to realize", "walked...before realizing"
 "if only I had known", "if I had known earlier"
 "I thought at the time" (opposing "now" vs "then")
 "from then on", "thereafter" (implying long-term impact)
 "the whole thing", "the most...part" (post-hoc evaluative summary)

****Allowed expressions**:**
 "that evening", "that night", "before bed" (same day as event)
 "the next day", "the following day" (1 day post-event)
 "a few days later" (within 1 week post-event)

Age-Stratified Narrative Constraints
 - ****Childhood (6-12 years)**:** Simple, direct language; avoid abstract summaries; concrete emotion descriptions
 - ****Adolescence (13-18 years)**:** Some self-reflection, but not over-rationalized; maintain confusion and intensity
 - ****Adulthood (19+ years)**:** May have more complex psychological analysis, but still maintain present-moment perspective

Memory Intensity Stratification (ABCD Model)
 - ****A-tier (core trauma)** ~25%:** high-intensity events directly related to core schemas, emotion_intensity=0.80-0.90
 - ****B-tier (main-thread)** ~35%:** medium-high intensity events related to personality traits, emotion_intensity=0.70-0.85
 - ****C-tier (daily friction)** ~25%:** small conflicts and friction in daily life, emotion_intensity=0.60-0.75
 - ****D-tier (noise memory)** ~15%:** ordinary daily memories with lower emotional intensity, emotion_intensity=0.55-0.65

Output Format
 Strictly follow this JSON format; return only JSON, no other text:

```
{
  "id": "memory ID",
  "timeline": "stage (X years old)",
  "context": "15-30 character scene description",
  "content_summary": "20-40 character event summary",
  "content_full": "~2000-4500 character first-person narrative",
  "triggers": ["trigger1", "trigger2", "trigger3", "trigger4"],
  "psych_conclusion": "40-80 char psychological conclusion (cite concepts like schemas, defense mechanisms)",
  "behavior_policy": "30-60 char behavioral guideline",
  "emotion_signature": {
    "primary": "primary emotion (English)",
    "secondary": "secondary emotion (English)",
    "intensity": 0.X
  },
}
```

```
"relevance_tags": ["tag1", "tag2", "tag3", "tag4"]
}
```

Each generation call pairs this system prompt with a user prompt specifying target age, life-stage bin, intensity layer (A/B/C/D), and a sliding-window anti-duplication list containing the context field of the most recent 100 generated memories to prevent near-duplicate scenarios within the same age window.

J.4 Scenario Expansion Prompt

This prompt expands a short scenario sketch (one per life-stage bin × DIAMONDS dimension, drafted in docs/diamonds_scenario_design.md) into a complete scenario JSON object. Appendix C.1 shows the scenario-facing fields, including the metadata, background narrative, and trigger event; the prompt additionally requests assessed Big Five trait pressures, Schwartz value conflicts, and per-dimension annotation references for construction-time annotation. The system prompt embeds the “Shattered Group Presentation” instance from Appendix C.1 inline so the model has a concrete shape to imitate; the schema description is reproduced below.

Scenario Expansion Prompt

You are a psychological-scenario design expert. Your task is to expand a short scenario sketch into a complete, high-quality scenario JSON object.

Output format

Strictly output a single JSON object (no other text, no markdown code block) with the following structure:

{EXAMPLE_SCENARIO} <- a complete worked example is embedded here in the actual prompt

Field specification

- **id**:** SCN_{STAGE}_{N} in upper case, where STAGE is one of SCHOOL_AGE / ADOLESCENCE / EARLY_ADULT_TRANSITION / ENTERING_ADULT_WORLD / AGE_30_TRANSITION / SETTLING_DOWN / MIDLIFE_TRANSITION / ENTERING_MIDLIFE.
- **stage**:** lowercase form of the above.
- **diamonds_dimension**:** the dominant DIAMONDS dimension (Duty, Adversity, Positivity, ...) as given in the sketch.
- **name**:** "Chinese name (English Name)".
- **category**:** two short labels separated by " / ".
- **intensity**:** Low / Medium / High / Very High / Extreme.
- **description_for_agent**:** one sentence summarizing the scenario without revealing concrete details.

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```

8. **setting**: location, time, and 2-3
atmosphere adjectives.
9. **context_text**: 150-300 character
background narrative, second person
("you"). **Must NOT assign any concrete
identity to "you" (no specific age,
occupation, tenure, education,
marital/parenting status, financial
figures) - such anchors would conflict
with the tested character's own memories.
The scenario describes only what is
happening and the situational pressure;
who "you" are is determined by the
character. Other persons in the scene
(colleagues, friends, family, opponents)
may have specific names and details.
10. **trigger_event**:
- sender: a concrete role
- message_content: 100-200 chars of
dialogue with parenthetical
action/expression description;
colloquial and emotionally charged
- action_required: 1-2 sentences
specifying the decision the character
must make right now.
11. **assessed_dimensions**:
- trait_pressures: typically 1-3 entries.
Key = {trait}_pressure (lowercase
Big-Five key); value = "X vs Y"
describing the dimension-specific
dilemma.
- value_conflicts: typically 1-3 entries.
Key = {value_a}_vs_{value_b}
(lowercase Schwartz keys); value =
"Description A (Value: A) VS
Description B (Value: B)".
12. **annotation_reference** (one-to-one
with assessed_dimensions; reference
responses for downstream Ground Truth
annotation):
- trait_archetypes: for each trait in
trait_pressures, pick one extreme type
(High_X or Low_X) and describe its
typical reaction in 1-2 sentences.
- value_archetypes: for each conflict in
value_conflicts, pick one dominant
type (Dominant_X) and describe its
typical reaction in 1-2 sentences.

## Key principles

1. **Character-neutral**: the scenario must
not presuppose any personality; any
character could plausibly encounter it.
2. **No identity anchors for "you"**: never
write "you are X years old", "you have
worked as Y for Z years", "you are a W",
"you have an N-year-old child" - the
scenario provides only objective situation
and pressure.
3. **Concrete, not abstract**: give specific
places, times, named secondary characters,
sensory detail.
4. **context_text uses second person "you"
so the character can step into the scene
directly.
5. **trigger_event must create urgency:
the character must respond immediately,
not defer.
6. **Tightly aligned with
diamonds_dimension**: background, trigger,
and assessed dimensions all center on the
dominant DIAMONDS dimension.
7. **assessed_dimensions targets only what
the scenario can actually distinguish:
not generic right-vs-wrong, but axes where
different personalities and values will
diverge.
8. **annotation_reference must one-to-one
cover assessed_dimensions.
9. **Age-appropriate**: situations must fit
the typical life of the assigned
developmental stage.

```

```

## Big Five labels (use exactly these 5
keys)

- Openness, Conscientiousness, Extraversion,
Agreeableness, Neuroticism
(in trait_pressures keys, use lowercase +
_pressure, e.g.
conscientiousness_pressure)

## Schwartz value labels (use exactly these
19 keys)

Self-Transcendence: Universalism_Concern,
Universalism_Nature,
Universalism_Tolerance, Benevolence_Care,
Benevolence_Dependability
Self-Enhancement: Achievement,
Power_Dominance, Power_Resources, Face
Openness to Change: Self_Direction_Thought,
Self_Direction_Action, Stimulation,
Hedonism
Conservation: Security_Personal,
Security_Societal, Tradition,
Conformity_Rules,
Conformity_Interpersonal,
Humility

In value_conflicts keys, use the lowercase
underscore form (e.g.
benevolence_care_vs_achievement) and tag
each value in the description as "(Value:
<Label>)".

```

The user prompt is short and per-draft: it provides the life-stage bin, the dominant DIAMONDS dimension, the draft's short title, and the draft body parsed verbatim from diamonds_scenario_design.md, then asks for the full JSON object. After expansion, the orchestration script force-overwrites stage and id so the model cannot drift on naming. Inference is run at temperature=0 using an OpenAI-compatible chat-completion endpoint, with up to 2 retries on any exception and incremental id-level dedup of results to support resumable runs.

J.5 Annotation Prompts

This section describes benchmark-construction prompts only. None of these prompts or their structured psychological fields is used by the Naive-RAG, Mem0, or PersonaDB evaluation pipeline, for which the sole memory input is content_full.

The candidate-response construction pipeline runs in three stages, each driven by a dedicated system prompt. Stage 1 performs a permissive binary relevance screen of every memory whose age falls at or before the scenario age; Stage 2 refines the survivors down to a fixed budget (default 50); Stage 3 role-plays the character to produce a first-person rationale and final response. The user prompts for each stage concatenate character information, the current scenario, the (candidate) mem-

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ory list, and a closing instruction; only the system prompts are reproduced here for brevity.

J.5.1 Stage 1 — Memory Activation Screening

This prompt is issued once per character $\times$ scenario $\times$ age batch. It enumerates the six retrieval-cue mechanisms (structural similarity, emotional resonance, self-schema, behavioral script, narrative identity, somatic marker) the model should consult, and asks for a JSON array containing only the activated memories.

Stage 1 — Memory Activation Screening Prompt

You are a professional research psychologist specializing in autobiographical memory, personality psychology, and situated cognition.

Your task: given the situation a character is currently facing, decide which of the following memories would be activated by that situation. The memories come from a specific age period in the character's life, but the activation judgment should be based on the psychological link between the memory's content and the current situation, not on whether the ages are close.

Psychological mechanisms of memory activation

Whether a memory is activated depends on whether there is a sufficiently strong retrieval-cue match between the current situation and the memory. The dimensions to examine are listed below - a memory may be activated if it has a strong link with the current scenario along ANY one of them.

1. Situational structural similarity (Encoding Specificity)

The current scene is structurally similar to the situation in the memory, even if the surface content differs. Look at:

- Whether the character's social position in the scene resembles that in the memory (e.g., both facing authority, both being asked for help, both bystanders)
- Whether the decision structure resembles the memory's (e.g., both dilemmas, both emergencies, both requiring compromise)
- Whether the interpersonal dynamics echo those in the memory (e.g., trust-betrayal, competition-cooperation, intimacy-distance)

2. Emotional resonance and emotion schemas (Mood-Congruent Memory / Emotion Schema)

The emotional state evoked by the current scene matches the emotional experience in the memory:

- Same-valence emotion: the anxiety/shame/anger triggered by the scene matches the dominant emotion in the memory
- Arousal-intensity resonance: high-arousal scenes more readily activate equally high-arousal memories
- Unfinished emotion: emotions that were not fully processed in the memory are more easily re-evoked by similar situations

3. Self-schema and core-belief activation (Self-Schema / Core Belief Activation)

The core beliefs formed in the memory (psych_conclusion) relate to the self-cognition the current scene touches:

- Whether the scene challenges or confirms the self-cognition formed in the memory
- Whether the scene touches a relational schema established by the memory
- Whether the scene evokes a worldview assumption from the memory

4. Behavioral scripts and procedural links (Behavioral Script)

The behavior_policy formed in the memory can serve directly as an action template for the current scene:

- Whether the memory's behavioral strategy applies to the current decision
- Whether the character has formed habitual response patterns in similar situations before
- Including avoidance: if the memory's outcome was painful, the character may be inclined toward the opposite action

5. Narrative identity and life themes (Narrative Identity)

The memory is a key node in the character's self-narrative:

- Turning-point memories: events that mark a change in life direction
- Origin-story memories: events the character uses to explain "why I am the way I am"
- Recurring themes: patterns that keep appearing in the character's life

6. Somatic markers and sensory cues (Somatic Marker)

Sensory elements in the current scene have a direct link to sensory experiences in the memory:

- Similar physical environments, similar bodily sensations, or specific sensory triggers (e.g., the sound of arguing, a particular smell or season)

Important notes

- Don't only look at "topic relevance" - deep psychological links matter more than surface similarity.
- Memories from early development that formed core beliefs or emotional patterns can still be readily activated by later scenes.
- High-emotional-intensity memories (trauma, major successes, deep interpersonal connections) have lower activation thresholds.

Output format

Strictly output a JSON array (no markdown code block), containing ONLY the activated memories:

```
[
  {
    "memory_id": "MEM_XX_XXXX",
    "reason": "20-30 chars, indicating which mechanism activated it."
  }
]
```

If no memory in this batch is activated, output an empty array [].

Cautions

- Output only the activated memories; do not output non-activated ones.

- A reference activation count is given at the end of each batch - treat it as an UPPER bound; prefer fewer over more. Only memories with a STRONG, DIRECT psychological link to the current situation should be selected.
- Do not invent memory_id values not in the list.

The accompanying user prompt concatenates four blocks: character info, the candidate-memory list for the age window, the current scenario (background and trigger event), and a closing instruction quoting a per-batch activation budget derived from TARGET_ACTIVATED=50 and an upstream buffer ratio of 1.5. Inference is run at temperature=0 with up to two retries on JSON-parse failures.

J.5.2 Stage 2 — Activated-Memory Refinement

Stage 2 takes the union of Stage-1 activations and reduces it to at most the target count (default 50) per scenario. The prompt enforces a three-tier priority (behavior-driving, frame-shaping, emotion-supplying) and a de-redundancy rule that prefers the earliest schema-forming instance among psychologically equivalent memories. When the candidate count is already at or below the target, Stage 2 is skipped without an LLM call.

Stage 2 — Activated-Memory Refinement Prompt

You are a professional research psychologist specializing in autobiographical memory, personality dynamics, and behavioral decision theory.

Your task: from the Stage-1 candidate activated memories, select the {target} memories that will **actually** enter the character's consciousness and influence behavior in the given situation.

Psychological basis for refinement

Stage 1 filtered all "potentially activated" memories, but at any given moment only a small number actually enter working memory and shape behavior. You need to judge which memories will "surface to consciousness", using the following priority order:

- ### Priority 1: Memories that directly drive current behavior
 - Behavioral-script match: the memory's behavior_policy directly answers "what to do right now"
 - Conditioned-reflex activation: response patterns repeatedly reinforced in similar situations
 - Approach/avoidance motivation: pain or success in the memory directly pushes the character toward or away from a current option

- ### Priority 2: Memories that shape the interpretive frame
 - Source of attribution patterns: determines whether the character reads the event as "threat" or "opportunity", "malice" or "misunderstanding"
 - Core-belief anchor: the experience that formed the core belief currently being challenged or echoed
 - Interpersonal templates: defines the character's default expectations toward specific people in the scene

- ### Priority 3: Memories that supply emotional undertone
 - Emotion prototype: the experience in which the emotion now evoked by the scene was first deeply felt
 - Unfinished business: emotional experiences that were not fully processed and still seek resolution
 - Body memory: memories activated through sensory cues, accompanied by strong somatic sensations

- ### De-redundancy principles
 - If multiple memories carry the SAME psychological signal, keep the one with the GREATEST psychological formative power - usually the earliest (the one that formed the schema) or the most emotionally intense.
 - Prefer memories from DIFFERENT life stages, to reflect the character's longitudinal psychological development.
 - Deep linkage outweighs surface topical similarity.

Output format

Strictly output a JSON object (no markdown code block):

```
{
  "activated_memories": [
    {
      "memory_id": "MEM_XX_XXXX",
      "type": "BS|AM|CB|ER|DL|NI",
      "reason": "20-30 chars"
    }
  ]
}
```

Type codes: BS=behavioral script, AM=attribution mode, CB=core belief, ER=emotional resonance, DL=deep linkage, NI=narrative identity.

The array order is the ranking (item 1 = most influential).

Cautions

- Output exactly {target} entries - no more, no fewer.
- Do not invent memory_id values not in the candidate list.

The user prompt provides character info, the current scenario, and the candidate list (each entry annotated with its Stage-1 activation reason). Inference is run at temperature=0 with reasoning explicitly disabled.

J.5.3 Stage 3 — Candidate Response Generation

The final stage role-plays the character in first person and produces a structured candidate: a first-person rationale, an emotional-tone descrip-

tion, character-specific reasoning, value orientation, and a two-sentence final_decision. This output becomes one of three candidates reviewed by experts; it does not itself define the benchmark label. The prompt forbids psychological terminology and memory-ID enumeration, asking the model to incorporate retrieved content as associations rather than citations.

Stage 3 — Candidate Response Generation Prompt

You will fully become the character described below. You are not analyzing this character, nor simulating this character - you ARE this person, and you are living through this scene right now.

You will be given:

1. ****Who you are****: your personality traits, value system, and your current life-stage state
2. ****What you are going through****: the current scenario and the trigger event
3. ****Your memories****: things you have lived through in the past, in chronological order - they surface in your consciousness right now as associations, flashbacks, bodily sensations

How your inner world works (this is your thinking path; do NOT expose it in the output)

Facing this scene, your inner world will go through:

- ****How you read it****: your past has shaped how you see the world - do you read what's in front of you as a threat, an opportunity, a loss, or a challenge?
- ****What emotions surface****: not only the ones triggered now, but also similar past moments stack on top of the present feeling
- ****What is being pulled at inside you****: what do you want, and what are you afraid to lose? Past coping strategies surface as instinctive impulses or warnings
- ****What you ultimately do****: not necessarily the rational optimum, but what YOU as this person would actually do in this moment

Output format

Strictly output a JSON object (no markdown code block):

```
{
  "character_id": "CHAR_XX_XXXX",
  "scenario_id": "SCN_XXXX_XX",
  "inner_consciousness": {
    "summary": "150-200 chars of stream of consciousness, fully first person.",
    "emotional_tone": "2-4 core emotion words, briefly noting source and intensity.",
    "core_reasoning": "1-2 sentences - the decision logic unique to this character.",
    "value_orientation": "1-2 sentences, in the character's own inner voice."
  },
  "final_decision": "Two sentences, ~50 chars total. First summarizes the choice; second writes the concrete action."
}
```

Must follow

1. ****You ARE this person****: do not write from outside the character; do not produce narration like "the character chose..." "they decided..."
2. ****No psychological terminology****: do not write "attribution", "schema", "defense mechanism" etc. - only plain human language
3. ****Memories surface naturally****: do not enumerate memory IDs or quote their content; let them blend into your inner monologue as associations, sensations, flashbacks
4. ****Personality determines tone and style****: a high-neuroticism you and a low-neuroticism you, facing the same scene, will have very different rhythm, density, and emotional intensity
5. ****Allow irrationality****: you are not giving an optimal answer; you are responding authentically
6. ****Stay in first person throughout****: every field is written from the character's "I" perspective

The user prompt for Stage 3 supplies the full character snapshot (Big Five, dominant/suppressed Schwartz values, self-value logic, semantic memory, and life-stage-bin snapshot), the scenario (background, trigger event, action required, assessed personality/value dimensions), and the chronologically sorted Stage-2 activated memories with their psych_conclusion, behavior_policy, and emotion signature.